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Risk Budgeting Allocation for Dynamic Risk Measures

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Abstract. We define and develop an approach for risk budgeting allocation—a risk diversification portfolio strategy—where risk is measured using a dynamic time-consistent risk measure. For this, we introduce a notion of dynamic risk contributions that generalize the classical Euler contributions, which allows us to obtain dynamic risk contributions in a recursive manner. We prove that for the class of coherent dynamic distortion risk measures, the risk allocation problem may be recast as a sequence of strictly convex optimization problems. Moreover, we show that self-financing dynamic risk budgeting strategies with initial wealth of one are scaled versions of the solution of the sequence of convex optimization problems. Furthermore, we develop an actor-critic approach, leveraging the elicitability of dynamic risk measures, to solve for risk budgeting strategies using deep learning.

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Keywords: dynamic risk measures • portfolio allocation • risk parity • elicitability • deep learning

1. Introduction

The "risk parity" portfolio was pioneered by Bridgewater Associates when in 1996, it launched the *All Weather* asset allocation strategy—a portfolio strategy withstanding all weathers—although the term risk parity was only coined in 2005 in the white paper by Qian (2005). Risk parity originated from the desire of a diversified portfolio and the realization that an equally weighted portfolio is diversified in asset allocation but not in the extent in which each asset contributes to the overall portfolio risk (Qian 2011). Emphasized by the 2008 financial crisis, the call for "maximally" diversifying a portfolio's risk was born; see, for example, Choueifaty and Coignard (2008). Risk parity enjoys widespread popularity in industry as numerous portfolio (performance) comparison studies illustrate; see, for example, Chaves et al. (2011), Lee (2011), and Asness et al. (2012). An early mathematical formalization of risk parity strategies can be found in Maillard et al. (2010) and Roncalli (2013).

Risk parity strategies and broader risk budgeting strategies are portfolio allocations where the contribution of each asset to the overall portfolio risk is prespecified (e.g., for risk parity, each asset contributes equally to the portfolio risk). Thus, central to risk budgeting is the way that the risk of a portfolio is quantified. Although most of the extant literature measures risk using the portfolio variance and further restricts to assets that follow multivariate Gaussian distributions, recent works relax these assumptions. Bruder et al. (2016) and Jurczenko and Teiletche (2019) study the Expected Shortfall (ES; also called Conditional Value-at-Risk) risk measure under the assumption that assets are multivariate Gaussian distributed, resulting in explicit formulas for risk contributions. Further works on risk budgeting include those by Ji and Lejeune (2018), who utilize the downside risk measure; Bellini et al. (2021), who consider expectile risk measures; Anis and Kwon (2022), who incorporate asset selection; and da Costa et al. (2023), who propose algorithms based on the cutting planes methodology to calculate risk budgeting strategies for coherent risk measures. Haugh et al. (2017) combine risk budgeting of (overlapping) groups of assets with simultaneously maximizing return and minimizing risk. Variations of risk budgeting portfolio strategies are considered by Bai et al. (2016), who propose

alternative optimization problems to solve for risk parity portfolios. Meucci et al. (2015) and Roncalli and Weisang (2016) construct risk factor budgeting portfolios that are portfolios where each (uncorrelated) factor, rather than asset, contributes equally to the portfolio variance. Lassance et al. (2022) continue this line of work by including independent component analysis.

None of these works, however, address the dynamic nature of investments (i.e., that portfolio strategies are typically holistically considered over a time horizon larger than one period); we henceforth refer to the one-period setting as the "static" setting. In this paper, we develop a dynamic setting in which an investor trades over a finite time horizon using a self-financing risk budgeting strategy. Specifically, the investor aims to create a portfolio strategy such that at every decision point, each asset contributes a prespecified percentage to the future risk of the portfolio. This means that the investor's problem is a multiperiod decision problem. Whenever decisions occur over multiple periods, the investor's "optimal" choices should be consistent over time. This can be achieved, for example, by optimizing a time-consistent criterion.[1] Indeed, when decisions are stemming from a time-inconsistent objective (e.g., a static risk measure), the "optimal" decision at a future point in time and state may not be optimal when one arrives at that future point in time in that very state (Bielecki et al. 2018). This is in contrast to time-consistent objectives (e.g., dynamic time-consistent risk measures (DRMs)), which result in decisions that are consistent across time and state. Thus, we consider dynamic time-consistent risk measures to evaluate the risk of a portfolio strategy.

Dynamic time-consistent risk measures have been extensively studied to evaluate the risk of a sequence of random costs, such as the profit and loss of a portfolio strategy; indicatively, see Cheridito et al. (2006), Ruszczyński (2010), Bielecki et al. (2022, 2024), Coache et al. (2023), and Moresco et al. (2024). For dynamic risk budgeting, however, we further require the allocations of the dynamic risk to each asset and each time point, a topic whose literature is sparse. An early work on allocations of coherent dynamic risk measures is Cherny (2009), and for Backward Stochastic Differential Equation (BSDE)-based dynamic time-consistent risk measures, we refer to Kromer and Overbeck (2014, 2017) and Mastrogiacomo and Rosazza-Gianin (2024). Related but conceptually different is the work of Schilling et al. (2020), who axiomatically study how to decompose a risk dynamically. Although working in a dynamic setting, their risk is the portfolio loss itself and not a dynamic risk measure applied to it.

In this work, we consider the class of dynamic time-consistent risk measures that arise from conditional one-step distortion risk measures. A case in point is the ES whose security level may depend on the investor's wealth or asset price. For this class, we define their dynamic risk contributions via Gâteaux derivatives and derive explicit formulas. Although most of our results hold for conditional distortion risk measures, we focus on the subset of conditional coherent distortion risk measures as defining risk allocations for noncoherent risk measures provides an "incentive for infinite fragmentation of portfolios" (Tsanakas 2009, p. 276). In the static setting, Gâteaux derivatives enjoy a long history as risk contributions also in connection to cooperative game theory. We provide a detailed literature review in Section 3. With this definition of dynamic risk contributions at hand, we define a dynamic risk budgeting portfolio as a strategy whose risk contributions at each point in time are a predefined percentage of the future risk of the strategy. We prove, under mild conditions, that a self-financing dynamic risk budgeting strategy with initial wealth of one is a scaled version of the solution of a sequence of strictly convex optimization problems. Furthermore, we develop an actor-critic approach to solve the sequence of optimization problems using deep learning techniques and provide illustrative examples.

This manuscript is organized as follows. Section 2 introduces dynamic time-consistent risk measures, and in Section 2.2, we apply a dynamic time-consistent risk measure to a self-financing strategy and derive a recursive representation. In Section 3, we define dynamic risk contributions via the Gâteaux derivative and derive explicit formulas for the class of dynamic distortion risk measures. Section 4 is devoted to dynamic risk budgeting portfolio strategies, and we show in Theorem 3 that a self-financing dynamic risk budgeting strategy with initial wealth of one can be obtained as a scaled version of the solution to a collection of strictly convex optimization problems. Section 5 discusses how to solve this family of optimization problems using neural networks (NNs) leveraging elicitability of conditional risk measures (Section 5.2). Illustrations of risk budgeting strategies are provided in Section 6. Delegated to the appendix are auxiliary definitions and results (Appendix A), additional technical proofs (Appendix B), elaborations on conditional elicitability (Appendix C), and details on the numerical implementation (Appendix D).

2. Dynamic Risk Assessment

We work on a filtered and completed probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in\overline{\mathcal{T}}}, \mathbb{P})$, where $\overline{\mathcal{T}} := \{0, 1, \dots, T+1\}$, and $T \in \mathbb{N}$ is a known and finite time horizon. The information available to the investor is encapsulated in the filtration $(\mathcal{F}_t)_{t\in\overline{\mathcal{T}}}$, and we assume that $\mathcal{F}_0 = \{\emptyset, \Omega\}$ is the trivial σ-algebra and simply write $\mathbb{E}[\cdot] := \mathbb{E}[\cdot|\mathcal{F}_0]$. We further denote the spaces of square-integrable random variables (rvs) and sequences by $\mathcal{Z} := \{Z \in \mathcal{F} : \mathbb{E}[Z^2] < +\infty\}$, $\mathcal{Z}_t := \{Z_t \in \mathcal{Z} : Z_t \in \mathcal{F}_t\}$, and $\mathcal{Z}_{t:T+1} := \{(Z_t, Z_{t+1}, \dots, Z_{T+1}) \in \mathcal{Z}_t \times \mathcal{Z}_{t+1} \times \cdots \times \mathcal{Z}_{T+1}\}$ for all $t \in \overline{\mathcal{T}}$. Similarly, we define the spaces of n-dimensional random vectors and sequences

by $\mathcal{Z} := \{\mathbf{Z} = (Z_1, \dots, Z_n) : Z_i \in \mathcal{Z},\ \forall i = 1, \dots, n\}$, $\mathcal{Z}_t := \{\mathbf{Z}_t \in \mathcal{Z} : \mathbf{Z}_t \in \mathcal{F}_t\}$, and $\mathcal{Z}_{t:T+1} := \{(\mathbf{Z}_t, \mathbf{Z}_{t+1}, \dots, \mathbf{Z}_{T+1}) \in \mathcal{Z}_t \times \mathcal{Z}_{t+1} \times \cdots \times \mathcal{Z}_{T+1}\}$ for all $t \in \overline{\mathcal{T}}$. We further define $\mathcal{L}^\infty := \mathcal{L}^\infty(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{L}^\infty_t := \mathcal{L}^\infty(\Omega, \mathcal{F}_t, \mathbb{P})$ for all $t \in \overline{\mathcal{T}}$. Unless otherwise stated, all (in-)equalities of random vectors are to be understood component wise and in a $\mathbb{P}$-almost sure (a.s.) sense.

2.1. Dynamic Risk Measures

The agent assesses the risk associated with a trading strategy by a dynamic time-consistent risk measure, which includes families of conditional risk measures that satisfy the property of strong time-consistency; see Definition 2 below. We adopt the setting of Cheridito et al. (2006) and Ruszczyński (2010) for dynamic risk measures and refer the interested reader to those works and the reference therein.

Definition 1 (Dynamic Risk Measures). A dynamic risk measure on $\overline{\mathcal{T}}$ is a family $\{\rho_{t,T+1}\}_{t\in\overline{\mathcal{T}}}$, where for each $t \in \overline{\mathcal{T}}$, the conditional risk measure is a mapping $\rho_{t,T+1} : \mathcal{Z}_{t:T+1} \to \mathcal{Z}_t$. We say that a dynamic risk measure possesses one of the following properties if for all $t \in \overline{\mathcal{T}}$:

i. Normalization. $\rho_{t,T+1}(0, \dots, 0) = 0$.

ii. Monotonicity. $\rho_{t,T+1}(Z_{t:T+1}) \le \rho_{t,T+1}(Y_{t:T+1})$ for all $Z_{t:T+1}, Y_{t:T+1} \in \mathcal{Z}_{t:T+1}$ with $Z_{t:T+1} \le Y_{t:T+1}$.

iii. Translation invariance. $\rho_{t,T+1}(Z_{t:T+1}) = Z_t + \rho_{t,T+1}(0, Z_{t+1}, \dots, Z_{T+1})$ for all $Z_{t:T+1} \in \mathcal{Z}_{t:T+1}$.

iv. Convexity. $\rho_{t,T+1}(\lambda Z_{t:T+1} + (1-\lambda) Y_{t:T+1}) \le \lambda \rho_{t,T+1}(Z_{t:T+1}) + (1-\lambda)\rho_{t,T+1}(Y_{t:T+1})$ for all $\lambda \in \mathcal{F}_t$ with $0 \le \lambda \le 1$ and $Y_{t:T+1}, Z_{t:T+1} \in \mathcal{Z}_{t:T+1}$.

v. Positive homogeneity. $\rho_{t,T+1}(\lambda Z_{t:T+1}) = \lambda \rho_{t,T+1}(Z_{t:T+1})$ for all $\lambda \in \mathcal{L}^\infty_t$ with $\lambda > 0$ and $Z_{t:T+1} \in \mathcal{Z}_{t:T+1}$.

vi. Coherency. $\rho_{t,T+1}$ is monotone, translation invariant, convex, and positive homogeneous.

The mapping $\rho_{t,T+1}$ thus assesses the risk of the sequence $Z_{t:T+1} \in \mathcal{Z}_{t:T+1}$ viewed from time t by mapping it to an $\mathcal{F}_t$-measureable rv. The investor may view this as the $\mathcal{F}_t$-measurable quantity that they are willing to exchange in place of the sequence of future risks.

Next, we recall the notion of strong time-consistency, which for simplicity, we refer to as time-consistency. A dynamic risk measure is time-consistent if it compares risks in a consistent manner over time. Specifically, if the time s risk of one stochastic process is larger than another, then the former should also be riskier at an earlier time $t < s$ if the processes are a.s. equal at all times u satisfying $t \le u < s$. Thus, time-consistency is a property that results in optimal sequential decisions that are consistent when optimized at different points in time and that further leads to a dynamic programming principle for optimizing dynamic risk measures.

Definition 2 (Strong Time-Consistency (Cheridito et al. 2006)). A dynamic risk measure $\{\rho_{t,T+1}\}_{t\in\overline{\mathcal{T}}}$ is (strong) time-consistent if for all $Z_{t:T+1}, Y_{t:T+1} \in \mathcal{Z}_{t:T+1}$ that satisfy for some $s \in \{t, \dots, T+1\}$

$$Z_{t:s} = Y_{t:s} \quad \text{and} \quad \rho_{s,T+1}(Z_{s:T+1}) \le \rho_{s,T+1}(Y_{s:T+1}),$$

it holds that

$$\rho_{t,T+1}(Z_{t:T+1}) \le \rho_{t,T+1}(Y_{t:T+1})$$

and where $Z_{t:s} := (Z_t, \dots, Z_s, 0, \dots, 0)$ is understood as the projection of $Z_{t:T+1}$ onto $\mathcal{Z}_t \times \cdots \times \mathcal{Z}_s$.

Although not apparent at first, the theorem below shows that time-consistency creates a connection between dynamic risk measures and so-called one-step risk measures. In particular, the theorem states that a dynamic time-consistent risk measure induces a family of one-step risk measures, and conversely, any family of one-step risk measures defines a dynamic time-consistent risk measure. The following theorem was first established in Cheridito et al. (2006) and Ruszczyński (2010).

Theorem 1 (Recursive Relation). *Let $\{\rho_{t,T+1}\}_{t\in\overline{\mathcal{T}}}$ be a dynamic risk measure, which is monotone, normalized, and translation invariant. Then, $\{\rho_{t,T+1}\}_{t\in\overline{\mathcal{T}}}$ is time-consistent if and only if the following recursive representation holds*:

$$\rho_{t,T+1}(Z_t, \dots, Z_{T+1}) = Z_t + \rho_t(Z_{t+1} + \rho_{t+1}(Z_{t+2} + \cdots + \rho_{T-1}(Z_T + \rho_T(Z_{T+1})) \cdots)), \quad (1)$$

where the one-step risk measures $\{\rho_t\}_{t\in\mathcal{T}}$ are mappings $\rho_t : \mathcal{Z}_{t+1} \to \mathcal{Z}_t$, defined by $\rho_t(Z_{t+1}) := \rho_{t,T+1}(0, Z_{t+1}, 0, \dots, 0)$, for $Z_{t+1} \in \mathcal{Z}_{t+1}$. Moreover, the one-step risk measures are monotone, normalized, and translation invariant.

By Theorem 1, any family of mappings $\rho_t : \mathcal{Z}_{t+1} \to \mathcal{Z}_t$ that are monotone, normalized, and translation invariant for all $t \in \mathcal{T}$, $\mathcal{T} := \{0, 1, \dots, T\}$ gives rise to a dynamic time-consistent risk measure and vice versa. Thus, without loss of generalization, we make a slight abuse of terminology and call $\{\rho_t\}_{t\in\mathcal{T}}$ a dynamic time-consistent risk measure with representation (1).

For defining risk budgeting strategies, we further require the DRM to be positive homogeneous, convex, and thus, coherent. Throughout the remainder of the exposition, we focus on coherent distortion DRMs, which are a generalization of the class of distortion risk measures to the dynamic setting. Coherent distortion risk measures span the subclass of law-invariant coherent risk measures that are comonotonic additive (Kusuoka 2001). Dynamic distortion risk measures with deterministic distortion function have been considered in the work of Bielecki et al. (2024), who focus on coherent dynamic acceptability indices generated by families of distortion functions. Here, we allow the distortion function to be both time and state dependent, which differs from earlier works. For this, we first define for each $t \in \mathcal{T}$ the regular cumulative distribution function (cdf) of $Z \in \mathcal{Z}_{t+1}$ conditional on $\mathcal{F}_t$ as $F_{Z|\mathcal{F}_t}(z) := \mathbb{P}(Z \le z|\mathcal{F}_t)$ and the regular (left-)quantile function of Z conditional on $\mathcal{F}_t$ as $F^{-1}_{Z|\mathcal{F}_t}(u) := \inf\{y \in \mathbb{R} | F_{Z|\mathcal{F}_t}(u) \ge y\}$. Moreover,

define $U_{Z|\mathcal{F}_t} := F_{Z|\mathcal{F}_t}(Z)$, which is $\mathcal{F}_{t+1}$ measurable and conditional on $\mathcal{F}_t$, uniform distributed on (0,1).[2] We refer the reader to Definition A.1 in Appendix A for further discussion of one-step distortion risk measures and their representation via Choquet integrals.

Definition 3 (Coherent One-Step Distortion Risk Measures). For each $t \in \mathcal{T}$, let $\gamma_t : [0,1] \times \Omega \to \mathbb{R}_+$ be a (state-dependent) distortion weight function. This means that $\int_0^1 \gamma_t(u,\omega)\,du = 1$ and $\gamma_t(\cdot,\omega)$ is nondecreasing for all $\omega \in \Omega$ and that the rv $\gamma_t(u,\cdot) : \Omega \to \mathbb{R}_+$ is $\mathcal{F}_t$ measurable for every $u \in [0,1]$ and for all $t \in \mathcal{T}$. Then, the coherent distortion dynamic risk measure with weight functions $\{\gamma_t\}_{t\in\mathcal{T}}$ is the family $\{\rho_t\}_{t\in\mathcal{T}}$, where for each $t \in \mathcal{T}$ and $Z \in \mathcal{Z}_{t+1}$, the one-step risk measure ρ_t is defined as

$$\begin{aligned}\rho_t(Z) &:= \int_0^1 F^{-1}_{Z|\mathcal{F}_t}(u)\,\gamma_t(u)\,du = \mathbb{E}[Z\,\gamma_t(F_{Z|\mathcal{F}_t}(Z))|\mathcal{F}_t] \\ &= \mathbb{E}[Z\,\gamma_t(U_{Z|\mathcal{F}_t})|\mathcal{F}_t], \end{aligned} \tag{2}$$

where as usual, we suppress the dependence of γ_t on its second argument.

Allowing for a time- and state-dependent distortion weight function includes, for example, Expected Shortfall at level $\alpha_t \in \mathcal{F}_t$, $\alpha_t \in [0,1)$ $\mathrm{ES}_\alpha(\cdot)$, in which case $\gamma_t(u) = \frac{1}{1-\alpha_t}\mathbb{1}_{u\geq\alpha_t}$. The level α_t may, for example, increase as wealth decreases to express the fact that the investor becomes more risk averse if their wealth drops.

One-step distortion risk measures that are coherent give raise to coherent DRMs via representation (1). Throughout, we work with coherent distortion DRMs.

Assumption 1. *We assume that $\{\rho_t\}_{t\in\mathcal{T}}$ is a coherent distortion DRM with representation (2) that satisfies $\int_0^1 \gamma_t(u)^2\,du < C$ a.s. for some $C < +\infty$ for all $t \in \mathcal{T}$.*

The integrability condition on $\{\gamma_t\}_{t\in\mathcal{T}}$ guarantees by Lemma A.1 that any coherent one-step distortion risk measure is a mapping $\rho_t : \mathcal{Z}_{t+1} \to \mathcal{Z}_t$, $t \in \mathcal{T}/\{0\}$, and $\rho_0 : \mathcal{Z}_1 \to \mathbb{R}$.

2.2. Risk-to-Go of a Strategy

We denote by $\boldsymbol{X} = (\boldsymbol{X}_t)_{t\in\overline{\mathcal{T}}} \in \boldsymbol{\mathcal{Z}}_{0:T+1}$ the (strictly) positive a.s. n-dimensional price process of the universe of assets and consider an investor who invokes a long-only self-financing trading strategy and invests in all assets. We also denote by $\boldsymbol{\theta} = (\boldsymbol{\theta}_t)_{t\in\mathcal{T}} \in \mathcal{Z}_{0:T}$ a (not necessarily self-financing) strategy, where $\boldsymbol{\theta}_t = (\theta_{t,1},\dots,\theta_{t,n})$ is an n-dimensional, positive a.s. random vector representing the amount of shares invested in each asset at time t. In the sequel, we often use the "slice notation" $\boldsymbol{\theta}_{t_1:t_2} := (\boldsymbol{\theta}_{t_1}, \boldsymbol{\theta}_{t_1+1},\dots,\boldsymbol{\theta}_{t_2})$ for $0 \leq t_1 < t_2 \leq T$.

A strategy $\boldsymbol{\theta}$ induces a self-financing strategy $\boldsymbol{\vartheta} = (\boldsymbol{\vartheta}_t)_{t\in\mathcal{T}}$—referred to as the *induced self-financing strategy*—as follows:

$$\boldsymbol{\vartheta}_0 := \boldsymbol{\theta}_0 \quad \text{and} \quad \boldsymbol{\vartheta}_t := \frac{\boldsymbol{\vartheta}_{t-1}^\top \boldsymbol{X}_t}{\boldsymbol{\theta}_t^\top \boldsymbol{X}_t}\,\boldsymbol{\theta}_t, \quad \forall t \in \mathcal{T}/\{0\}.$$

Recall that the investor invokes a long-only strategy and invests in all assets; thus, $\theta_{t,i} > 0$, a.s., for all $i \in \mathcal{N} := \{1,\dots,n\}$ and $t \in \mathcal{T}$, and hence, $\boldsymbol{\vartheta}_{0:T}$ is well defined. The strategy $\boldsymbol{\vartheta}$ is self-financing (i.e., it satisfies $(\boldsymbol{\vartheta}_t - \boldsymbol{\vartheta}_{t-1})^\top \boldsymbol{X}_t = 0$ for all $t \in \mathcal{T}/\{0\}$). To simplify notation, we define the weight process $\boldsymbol{w}^{\boldsymbol{\theta}} = (\boldsymbol{w}_t^{\boldsymbol{\theta}})_{t\in\mathcal{T}}$:

$$w_t^{\boldsymbol{\theta}} := \frac{\boldsymbol{\theta}_t^\top \boldsymbol{X}_{t+1}}{\boldsymbol{\theta}_{t+1}^\top \boldsymbol{X}_{t+1}}, \quad \forall t \in \mathcal{T},$$

and we notice that

$$\boldsymbol{\vartheta}_t = \left(\prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}}\right)\boldsymbol{\theta}_t, \quad \forall t \in \mathcal{T}/\{0\}.$$

For each $t \in \mathcal{T}$, $w_t^{\boldsymbol{\theta}}$ is $\mathcal{F}_{t+1}$ measurable, and if the original strategy $\boldsymbol{\theta}_{0:T}$ is already self-financing, then $w_t^{\boldsymbol{\theta}} = 1$.

We denote the (negative) price increment by $\Delta\boldsymbol{X}_t := -(\boldsymbol{X}_{t+1} - \boldsymbol{X}_t)$. As the investor aims to invest in a risk budgeting portfolio, they only consider strategies that are long only and satisfy the following integrability conditions.

Definition 4. We denote the set of strategies satisfying integrability assumptions by

$$\mathcal{A} := \left\{ \boldsymbol{\theta}_{0:T} \in \mathcal{Z}_{0:T} | w_t^{\boldsymbol{\theta}} \in \mathcal{L}^\infty_{t+1},\ \forall t \in \mathcal{T}, \quad \text{and} \right.$$
$$\left. \sum_{t\in\mathcal{T},\, i\in\mathcal{N}} \mathbb{E}[(\theta_{t,i}\Delta X_{t,i})^2] < +\infty \right\}.$$

Furthermore, for any $c > 0$, we define

$$\mathcal{A}_c := \{\boldsymbol{\theta}_{0:T} \in \mathcal{A} | \boldsymbol{\theta}_{0:T} \geq c \ \text{ a.s.}\} \quad \text{and}$$
$$\mathcal{A}_0 := \{\boldsymbol{\theta}_{0:T} \in \mathcal{A} | \boldsymbol{\theta}_{0:T} > 0 \ \text{ a.s.}\}.$$

For any $c \geq 0$, by Lemma A.2, $\mathcal{A}_c$ is a convex set, and any induced self-financing strategy $\boldsymbol{\vartheta}$ belongs to $\mathcal{A}_0$. We further use the notation $\mathcal{A}_{c,t}$, $t \in \mathcal{T}$, to denote the sliced set of admissible strategies at time t; that is, $\mathcal{A}_{c,t} := \{\boldsymbol{\theta}_t | \boldsymbol{\theta}_{0:T} \in \mathcal{A}_c\}$ for each $t \in \mathcal{T}$. We use the terminology that the strategy $\boldsymbol{\theta}$ is *admissible* if there exists a $c \geq 0$ such that $\boldsymbol{\theta} \in \mathcal{A}_c$.

The investor assesses the risk of an admissible strategy $\boldsymbol{\theta}_{0:T} \in \mathcal{A}_c$ at time $t = 0$ with a coherent distortion DRM $\{\rho_t\}_{t\in\mathcal{T}}$ by

$$\begin{aligned}&\Re[\boldsymbol{\theta}_{0:T}] \\ &= \rho_0\left(\boldsymbol{\theta}_0^\top \Delta\boldsymbol{X}_0 + \rho_1\left(w_0^{\boldsymbol{\theta}}\,\boldsymbol{\theta}_1^\top \Delta\boldsymbol{X}_1 + \rho_2\left(w_0^{\boldsymbol{\theta}} w_1^{\boldsymbol{\theta}}\,\boldsymbol{\theta}_2^\top \Delta\boldsymbol{X}_2 + \cdots\right.\right.\right. \\ &\quad \left.\left.\left. + \rho_{T-1}\left(\prod_{s=0}^{T-2} w_s^{\boldsymbol{\theta}}\,\boldsymbol{\theta}_{T-1}^\top \Delta\boldsymbol{X}_{T-1} + \rho_T\left(\prod_{s'=0}^{T-1} w_{s'}^{\boldsymbol{\theta}}\,\boldsymbol{\theta}_T^\top \Delta\boldsymbol{X}_T\right)\right)\cdots\right)\right)\right) \\ &= \rho_0(\boldsymbol{\vartheta}_0^\top \Delta\boldsymbol{X}_0 + \rho_1(\boldsymbol{\vartheta}_1^\top \Delta\boldsymbol{X}_1 + \cdots + \rho_{T-1}(\boldsymbol{\vartheta}_{T-1}^\top \Delta\boldsymbol{X}_{T-1} \\ &\quad + \rho_T(\boldsymbol{\vartheta}_T^\top \Delta\boldsymbol{X}_T))\cdots)).\end{aligned}$$

Hence, $\Re[\boldsymbol{\theta}_{0:T}]$ is the dynamic risk of the induced self-financing strategy but parameterized by $\boldsymbol{\theta}_{0:T}$. We can

view the risk recursively by defining the risk-to-go process $(\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}])_{t\in\overline{\mathcal{T}}}$ via

$$\mathfrak{R}_{T+1} := 0 \quad \text{and} \tag{3a}$$

$$\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] := \rho_t(\boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + w_t^{\boldsymbol{\theta}} \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]), \quad \forall t \in \mathcal{T}. \tag{3b}$$

At time $t = 0$, it holds that $\mathfrak{R}_0[\boldsymbol{\theta}_{0:T}] = \mathfrak{R}[\boldsymbol{\theta}_{0:T}]$. By Lemma A.3, any admissible strategy $\boldsymbol{\theta} \in \mathcal{A}_c$, $c \geq 0$ (and thus, also any induced self-financing strategy) has finite risk $\mathfrak{R}[\boldsymbol{\theta}_{0:T}] < +\infty$ and satisfies $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] \in \mathcal{Z}_t$ for all $t \in \mathcal{T}/\{0\}$.

The risk-to-go evaluated along a self-financing strategy $\boldsymbol{\psi}_{0:T} \in \mathcal{Z}_{0:T}$ satisfies a slightly simpler recursion:

$$\mathfrak{R}_t[\boldsymbol{\psi}_{t:T}] := \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]|_{\boldsymbol{\theta}_{t:T}=\boldsymbol{\psi}_{t:T}} = \rho_t(\boldsymbol{\psi}_t^\top \Delta \boldsymbol{X}_t + \mathfrak{R}_{t+1}[\boldsymbol{\psi}_{t+1:T}]), \quad \forall t \in \mathcal{T},$$

as $w_t^{\boldsymbol{\psi}} = 1$ for all $t \in \mathcal{T}/\{0\}$ for self-financing strategies.

The next proposition connects the risk-to-go process of $\boldsymbol{\theta}_{0:T}$ with the risk-to-go process of its induced self-financing strategy $\boldsymbol{\vartheta}_{0:T}$.

Proposition 1. *Let $\boldsymbol{\theta}_{0:T}$ be an admissible strategy, and denote by $\boldsymbol{\vartheta}_{0:T}$ its induced self-financing strategy. Then, the following holds*:

$$\mathfrak{R}_0[\boldsymbol{\vartheta}_{0:T}] = \mathfrak{R}_0[\boldsymbol{\theta}_{0:T}] \quad \textit{and}$$

$$\mathfrak{R}_t[\boldsymbol{\vartheta}_{t:T}] = \left(\prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}}\right) \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}], \quad \forall t \in \mathcal{T}/\{0\}. \tag{4}$$

Proof. The equation for $t = 0$ follows by definition. To show the equalities for $t \in \mathcal{T}/\{0\}$, we proceed by induction starting backward in time. At time T, we use positive homogeneity of the conditional risk measures because by admissibility of $\boldsymbol{\theta}_{0:T}$, we have $0 < w_t^{\boldsymbol{\theta}} \in \mathcal{L}_{t+1}^\infty$ for all $t \in \mathcal{T}$. Thus,

$$\begin{aligned}\mathfrak{R}_T[\boldsymbol{\vartheta}_T] = \rho_T(\boldsymbol{\vartheta}_T^\top \Delta \boldsymbol{X}_T) &= \left(\prod_{s=0}^{T-1} w_s^{\boldsymbol{\theta}}\right) \rho_T(\boldsymbol{\theta}_T^\top \Delta \boldsymbol{X}_T) \\ &= \left(\prod_{s=0}^{T-1} w_s^{\boldsymbol{\theta}}\right) \mathfrak{R}_T[\boldsymbol{\theta}_T].\end{aligned}$$

Next, assume that Equation (4) holds for $t + 1$, and note that $w_s^{\boldsymbol{\theta}}$ is $\mathcal{F}_t$ measurable for all $0 \leq s < t$. Then, at time t, we have

$$\begin{aligned}\mathfrak{R}_t[\boldsymbol{\vartheta}_{t:T}] &= \rho_t(\boldsymbol{\vartheta}_t^\top \Delta \boldsymbol{X}_t + \mathfrak{R}_{t+1}[\boldsymbol{\vartheta}_{t+1:T}]) \\ &= \rho_t\left(\left(\prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}}\right) \boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + \left(\prod_{s=0}^{t} w_s^{\boldsymbol{\theta}}\right) \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]\right) \\ &= \left(\prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}}\right) \rho_t(\boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + w_t^{\boldsymbol{\theta}}\, \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]) \\ &= \left(\prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}}\right) \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}],\end{aligned}$$

where the first equality holds from the induction assumption. The second follows from positive homogeneity of the conditional risk measure and $0 < \prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}} \in \mathcal{L}_t^\infty$. The last equality follows from Equation (3). □

Here, as in the static risk budgeting problem, positive homogeneity of the risk measure plays a central role. Therefore, we next discuss the positive homogeneity of the risk-to-go process. For this, it is convenient to split the arguments of $\mathfrak{R}_t$ into two parts; specifically, we write $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] = \mathfrak{R}_t[(\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})]$ to emphasize the difference of the position at t, $\boldsymbol{\theta}_t$, and the remaining ones $\boldsymbol{\theta}_{t+1:T}$.

Proposition 2 (Positive Homogeneity of the Risk-to-Go Process). *Let $\boldsymbol{\theta}_{0:T}$ be an admissible strategy. Then, the risk-to-go process is positive homogeneous viewed as a function of $\boldsymbol{\theta}_t$ and also viewed as a function of $\boldsymbol{\theta}_{t:T}$; that is, for all $t \in \mathcal{T}$ and for all $a_t \in \mathcal{L}_t^\infty$, $a_t > 0$,*

$$a_t\, \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] = \mathfrak{R}_t[(a_t\, \boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})] = \mathfrak{R}_t[a_t\, (\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})]. \tag{5}$$

Proof. The first equality (i.e., positive homogeneity of $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]$ in $\boldsymbol{\theta}_t$) follows from representation (3), the linearity of $w_t^{\boldsymbol{\theta}}$ in $\boldsymbol{\theta}_t$ noting that $\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]$ does not depend on $\boldsymbol{\theta}_t$, and $\rho_t(\cdot)$ being positive homogeneous.

To see the second equality, we proceed by induction. First, $\mathfrak{R}_T[\boldsymbol{\theta}_T] = \rho_T(\boldsymbol{\theta}_T^\top \Delta \boldsymbol{X}_T)$ is positive homogeneous in $\boldsymbol{\theta}_T$. Next, $w_t^{\boldsymbol{\theta}}$ is invariant under scaling of both $\boldsymbol{\theta}_t$ and $\boldsymbol{\theta}_{t+1}$ (i.e., $w_t^{\boldsymbol{\theta}} = \frac{\boldsymbol{\theta}_t^\top \boldsymbol{X}_{t+1}}{\boldsymbol{\theta}_{t+1}^\top \boldsymbol{X}_{t+1}} = \frac{a_t\, \boldsymbol{\theta}_t^\top \boldsymbol{X}_{t+1}}{a_t\, \boldsymbol{\theta}_{t+1}^\top \boldsymbol{X}_{t+1}} = w_t^{\boldsymbol{\theta}'}$, $a_t \in \mathcal{L}_t^\infty$, $a_t > 0$, where $\boldsymbol{\theta}'_{0:T}$ is such that $\boldsymbol{\theta}'_t = a_t \boldsymbol{\theta}_t$, $\boldsymbol{\theta}'_{t+1} = a_t \boldsymbol{\theta}_{t+1}$, with all remaining $\boldsymbol{\theta}'_s = \boldsymbol{\theta}_s$ for $s \notin \{t, t+1\}$). Moreover, $\boldsymbol{\theta}'_{0:T}$ is admissible. Now, assume that the second equality in (5) holds for $t + 1$; then, we have

$$\begin{aligned}\mathfrak{R}_t[a_t(\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})] &= \rho_t(a_t\, \boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + w_t^{\boldsymbol{\theta}'}\, \mathfrak{R}_{t+1}[a_t\, (\boldsymbol{\theta}_{t+1:T})]) \\ &= \rho_t(a_t\, \boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + w_t^{\boldsymbol{\theta}}\, a_t\, \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]) \\ &= a_t\, \mathfrak{R}_t[\boldsymbol{\theta}_t; \boldsymbol{\theta}_{t+1:T}],\end{aligned}$$

where the first equality follows from (3), the second equality follows from the inductive assumption and that $w_t^{\boldsymbol{\theta}} = w_t^{\boldsymbol{\theta}'}$, and the last equality follows by positive homogeneity of $\rho_t(\cdot)$. □

We next discuss how to allocate the risk-to-go onto its components at each point in time.

3. Dynamic Risk Contributions

The literature on risk contribution—also called capital (cost) allocation—in the static setting is extensive. Approaches range from performance measurement (Tasche 1999) to cooperative game theory, including Aumann–Shapley allocation (see, e.g., Billera and Heath 1982 and Mirman and Tauman 1982 for early works on cost allocation and Denault 2001 for a risk management setting) and allocation in the fuzzy core (Tsanakas and Barnett 2003), as well as Gâteaux

derivatives and Euler allocations (Kalkbrener 2005). Using an axiomatic approach, Kalkbrener (2005) showed that for any positive homogeneous and subadditive static risk measure, the only linear and diversifying capital allocation rule is the Gâteaux derivative.

Here, we proceed in line with Kalkbrener (2005) by defining risk contributions as a subdifferential, specifically through the Gâteaux derivative. We note that in the case of coherent risk measures, the allocation defined via the Gâteaux derivatives is the same as the Aumann–Shapley allocation (Tsanakas 2009). An advantage of defining risk contributions through Gâteaux derivatives of a distortion risk measure is that they satisfy *full allocation*, the property that the sum of the risk contributions adds up to the total risk. In the sequel, we show that our dynamic risk contributions also satisfy a dynamic version of the full allocation property. When defining dynamic risk budgeting strategies, the full allocation property is imperative as it allows us to allocate the entire risk to its components.

As we work in a dynamic setting, at each time $t \in \mathcal{T}$, the investor faces the future risk of the induced self-financing strategy and aims to allocate the risk-to-go to each asset i. Thus, for each $t \in \mathcal{T}$, we define the risk contribution of asset i as the one-sided Gâteaux derivative of the risk-to-go processes $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]$ in direction $\theta_{t,i}$. This allows us to measure the degree to which the risk-to-go is impacted by the investor's position in the ith asset at time t. Moreover, as we show in Corollary 1, this approach allows for full allocation.

We start by recalling the definition of the one-sided Gâteaux derivative of a functional.

Definition 5. For a functional $F_t : \boldsymbol{\mathcal{Z}}_{t:T} \to \mathcal{Z}_t$, $t \in \mathcal{T}$, we denote by $\mathcal{D}_i^{\zeta} F_t$ its one-sided Gâteaux derivative of F_t to the ith component in direction $\zeta \in \mathcal{Z}_t$. That is, for $t \in \mathcal{T}$ and $\boldsymbol{Z}_{t:T} \in \boldsymbol{\mathcal{Z}}_{t:T}$,

$$\mathcal{D}_i^{\zeta} F_t[\boldsymbol{Z}_{t:T}] := \lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon}(F_t[\boldsymbol{Z}_{t:T} + \varepsilon \mathbf{1}_{t,i}\zeta] - F_t[\boldsymbol{Z}_{t:T}]),$$

where $(\mathbf{1}_{t,i})_{t \in \mathcal{T}}$ is the stochastic process taking a value of one in component i at time t and zero otherwise.

We consider the one-sided Gâteaux derivative as we work with long-only portfolios. Thus, when taking directional derivatives with respect to $\theta_{t,i}$, its perturbed values should also be positive, which is guaranteed with non-negative ε in Definition 5.

Definition 6. For each $t \in \mathcal{T}$, we define the risk contribution of the risk-to-go to the ith investment as

$$RC_{t,i}[\boldsymbol{\theta}_{t:T}] := \mathcal{D}_i^{\theta_{t,i}} \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}].$$

Note that the risk contributions $RC_{t,i}[\boldsymbol{\theta}_{t:T}]$ exist, are $\mathcal{F}_t$ measurable rvs, and satisfy $RC_{t,i}[\boldsymbol{\theta}_{t:T}] \in \mathcal{Z}_t$ for all $t \in \mathcal{T}$ and $i \in \mathcal{N}$; see Theorem 2 below.

The next result shows that the risk contributions are also positive homogeneous.

Proposition 3 (Positive Homogeneity of Risk Contributions). *The risk contributions of an admissible strategy $\boldsymbol{\theta}_{0:T}$ to the ith investment at time $t \in \mathcal{T}$ are positive homogeneous in the following way. For all $t \in \mathcal{T}$ and for all $a_t \in \mathcal{L}_t^{\infty}$, $a_t > 0$, we have that*

$$a_t\, RC_{t,i}[\boldsymbol{\theta}_{t:T}] = RC_{t,i}[(a_t\, \boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})] = RC_{t,i}[(a_t\, \boldsymbol{\theta}_{t:T})]. \quad (6)$$

Proof. This follows as $RC_{t,i}$ are the Gâteaux derivatives of a positive homogeneous function. For completeness, we provide a short proof. For any $t \in \mathcal{T}$ and $i \in \mathcal{N}$, we obtain from positive homogeneity of $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]$ in $\boldsymbol{\theta}_t$ (see Proposition 2) that

$$\begin{aligned} &RC_{t,i}[(a_t\, \boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})] \\ &= \lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon}(\mathfrak{R}_t[(a_t(\boldsymbol{\theta}_t + \varepsilon\, \boldsymbol{e}_i\, \theta_{t,i}), \boldsymbol{\theta}_{t+1:T})] - \mathfrak{R}_t[(a_t\, \boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})]) \\ &= \lim_{\varepsilon \downarrow 0} a_t\, \frac{1}{\varepsilon}(\mathfrak{R}_t[((\boldsymbol{\theta}_t + \varepsilon\, \boldsymbol{e}_i\, \theta_{t,i}), \boldsymbol{\theta}_{t+1:T})] - \mathfrak{R}_t[(\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})]) \\ &= a_t\, RC_{t,i}[\boldsymbol{\theta}_{t:T}], \end{aligned}$$

where $\boldsymbol{e}_i$ is the unit vector having a value of one at position i (and zero otherwise). The second equality in (6) follows via similar arguments using positive homogeneity of $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]$ in $\boldsymbol{\theta}_{t:T}$; see Proposition 2. □

By positive homogeneity of the risk-to-go process, we obtain for all $t \in \mathcal{T}$ a Euler-like theorem as stated in the next corollary, which guarantees full allocation.

Corollary 1 (Full Allocation). *Let $\{\rho_t\}_{t \in \mathcal{T}}$ be a coherent distortion DRM. Then, it holds for any admissible strategy $\boldsymbol{\theta}_{0:T}$ that*

$$\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] = \sum_{i \in \mathcal{N}} RC_{t,i}[\boldsymbol{\theta}_{t:T}], \quad a.s., \quad \forall t \in \mathcal{T}.$$

Proof. This proof follows similar steps as the classical Euler allocation. By positive homogeneity of the risk-to-go, Proposition 2, we have for all real numbers $a > 0$ that

$$a\, \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] = \mathfrak{R}_t[(a\, \boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})], \qquad t \in \mathcal{T}. \quad (7)$$

Next, viewing Equation (7) as a function of a, we take a one-sided partial derivative with respect to a of both sides of the above equation: that is,

$$\begin{aligned} \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] &= \lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon}(\mathfrak{R}_t[((a + \varepsilon)\, \boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})] - \mathfrak{R}_t[(a\, \boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})]) \\ &= \sum_{i \in \mathcal{N}} \mathcal{D}_i^{\theta_{t,i}} \mathfrak{R}_t[(a\, \boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T})]. \end{aligned}$$

Evaluating the above at $a = 1$ concludes the proof. □

In Tsanakas (2004), the author derives, in the static setting, a closed-form formula for the risk contributions

of distortion risk measures. Our next result extends this to the dynamic setting.

Theorem 2 (Risk Contributions). *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be a coherent distortion DRM with weight functions $\{\gamma_t\}_{t\in\mathcal{T}}$. Then, the risk contribution of an admissible strategy $\boldsymbol{\theta}_{0:T}$ to the ith investment at time $t\in\mathcal{T}$ is given by*

$$RC_{t,i}[\boldsymbol{\theta}_{t:T}] = \mathbb{E}\left[\theta_{t,i}\left(\Delta X_{t,i} + \frac{X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top \boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]\right)\gamma_t(U_t[\boldsymbol{\theta}_{t:T}])\middle|\mathcal{F}_t\right],$$

where $U_t[\boldsymbol{\theta}_{t:T}]$ is a uniform rv comonotonic to $\boldsymbol{\theta}_t^\top \Delta\boldsymbol{X}_t + w_t^{\boldsymbol{\theta}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]$. Furthermore, $RC_{t,i}[\boldsymbol{\theta}_{t:T}]\in\mathcal{Z}_t$ for all $t\in\mathcal{T}$ and $i\in\mathcal{N}$.

Proof. Using Tsanakas and Millossovich (2016, proposition 1) and Pesenti et al. (2021, proposition 3.2), we have that for $\boldsymbol{Y},\boldsymbol{Y}'\in\boldsymbol{\mathcal{Z}}_{t+1}$, a differentiable function $h:\mathbb{R}^d\to\mathbb{R}$, and a conditional distortion risk measure ρ_t, it holds that

$$\lim_{\varepsilon\downarrow 0}\frac{\rho_t(h(\boldsymbol{Y}'+\varepsilon\,\boldsymbol{e}_i\,Y_i)) - \rho_t(h(\boldsymbol{Y}'))}{\varepsilon} = \mathbb{E}\left[Y_i\,\frac{\partial}{\partial y_i}h(\boldsymbol{Y}')\,\gamma_t(U_{h(\boldsymbol{Y}')|\mathcal{F}_t})|\mathcal{F}_t\right], \tag{9}$$

where $U_{h(\boldsymbol{Y}')|\mathcal{F}_t}$ is a uniform rv that is comonotonic to the rv $h(\boldsymbol{Y}')$ conditional on the information $\mathcal{F}_t$; see also Equation (2). In Proposition A.3 in Appendix A, we provide an alternative proof of Equation (9).

Next, note that the risk contributions are

$$\begin{aligned}
RC_{t,i}[\boldsymbol{\theta}_{t:T}] &= \mathcal{D}_i^{\theta_{t,i}}\,\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]\\
&= \mathcal{D}_i^{\theta_{t,i}}\,\rho_t(\boldsymbol{\theta}_t^\top\Delta\boldsymbol{X}_t + w_t^{\boldsymbol{\theta}}\,\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}])\\
&= \mathcal{D}_i^{\theta_{t,i}}\,\rho_t\left(\sum_{i\in\mathcal{N}}\theta_{t,i}\left\{\Delta X_{t,i} + \frac{X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top\boldsymbol{X}_{t+1}}\,\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]\right\}\right).
\end{aligned}$$

Applying Equation (9) and noting that $\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]$ is a function of $\boldsymbol{\theta}_{t+1:T}$ only and not a function of $\theta_{t,1}$ conclude the first part of the statement. The last part follows from Lemma A.4. □

The next representation of the risk contributions illustrates that a decision at time t, via $\boldsymbol{\theta}_t$, cascades through time and impacts all later decision points. This is because restricting to self-financing strategies implies that the investor's future wealth and thus, also possible investment decisions depend on the current choice of $\boldsymbol{\theta}_t$. The proof is delegated to Appendix B.1.

Proposition 4 (Impact of a Decision). *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be a coherent distortion DRM with weight functions $\{\gamma_t\}_{t\in\mathcal{T}}$. Then, the risk contribution of an admissible strategy $\boldsymbol{\theta}_{0:T}$ to the ith investment at time $t\in\mathcal{T}$ may be written as*

$$\begin{aligned}
&RC_{t,i}[\boldsymbol{\theta}_{t:T}]\\
&= \mathbb{E}[\theta_{t,i}\,\Delta X_{t,i}\,\Gamma_t^{\boldsymbol{\theta}}|\mathcal{F}_t]\\
&\quad + \mathbb{E}\left[\frac{\theta_{t,i}\,X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top\boldsymbol{X}_{t+1}}\,(\boldsymbol{\theta}_{t+1}^\top\Delta\boldsymbol{X}_{t+1})\,\Gamma_t^{\boldsymbol{\theta}}\,\Gamma_{t+1}^{\boldsymbol{\theta}}\middle|\mathcal{F}_t\right]\\
&\quad + \mathbb{E}\left[\frac{\theta_{t,i}\,X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top\boldsymbol{X}_{t+1}}\,w_{t+1}^{\boldsymbol{\theta}}\,(\boldsymbol{\theta}_{t+2}^\top\Delta\boldsymbol{X}_{t+2})\,\Gamma_t^{\boldsymbol{\theta}}\,\Gamma_{t+1}^{\boldsymbol{\theta}}\,\Gamma_{t+2}^{\boldsymbol{\theta}}\middle|\mathcal{F}_t\right]\\
&\quad + \cdots + \mathbb{E}\left[\frac{\theta_{t,i}\,X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top\boldsymbol{X}_{t+1}}\,w_{t+1}^{\boldsymbol{\theta}}\cdots w_{T-1}^{\boldsymbol{\theta}}\,(\boldsymbol{\theta}_T^\top\Delta\boldsymbol{X}_T)\,\Gamma_t^{\boldsymbol{\theta}}\cdots\Gamma_T^{\boldsymbol{\theta}}\middle|\mathcal{F}_t\right],
\end{aligned}$$

where $\Gamma_s^{\boldsymbol{\theta}} := \gamma_s(U_s[\boldsymbol{\theta}_{s:T}])$, with $U_s[\boldsymbol{\theta}_{s:T}]$ as defined in Theorem 2 for all $s\in\mathcal{T}$.

From the above proposition, we can interpret the first expectation as the time t impact of $\theta_{t,i}$, the second expectation as the effect that the choice $\theta_{t,i}$ has at time $t+1$, and so on.

Note that the risk contribution of investment i at time t of a induced self-financing strategy $\boldsymbol{\vartheta}$ is

$$RC_{t,i}[\boldsymbol{\vartheta}_{t:T}] := D_i^{\theta_{t,i}}\,\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]\,|_{\boldsymbol{\theta}=\boldsymbol{\vartheta}}.$$

The next statement relates the risk contributions of a strategy with those of its induced self-financing strategy.

Proposition 5. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be a coherent distortion DRM with weight functions $\{\gamma_t\}_{t\in\mathcal{T}}$. Let $\boldsymbol{\theta}_{0:T}$ be admissible and $\boldsymbol{\vartheta}_{0:T}$ be its induced self-financing strategy. Then, the following holds:*

$$RC_{0,i}[\boldsymbol{\vartheta}_{0:T}] = RC_{0,i}[\boldsymbol{\theta}_{0:T}] \quad \textit{and}$$
$$RC_{t,i}[\boldsymbol{\vartheta}_{t:T}] = \left(\prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}}\right)RC_{t,i}[\boldsymbol{\theta}_{t:T}], \quad \forall t\in\mathcal{T}/\{0\}.$$

Proof. Case $t=0$ follows immediately from Proposition 1. For $t\in\mathcal{T}/\{0\}$, define the scalar rv $c_t^{\boldsymbol{\theta}} := \prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}} \in \mathcal{L}_t^\infty$, and recall that $\boldsymbol{\vartheta}_t = c_t^{\boldsymbol{\theta}}\,\boldsymbol{\theta}_t$. Then, we have that

$$\begin{aligned}
RC_{t,i}[\boldsymbol{\vartheta}_{t:T}] &= \mathcal{D}_i^{\theta_{t,i}}\,\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]\,|_{\boldsymbol{\theta}=\boldsymbol{\vartheta}}\\
\text{(by Theorem 2)} &= \mathbb{E}\left[\vartheta_{t,i}\left(\Delta X_{t,i} + \frac{X_{t+1,i}}{\boldsymbol{\vartheta}_{t+1}^\top\boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\vartheta}_{t+1:T}]\right)\right.\\
&\qquad\left.\gamma_t(U_t[\boldsymbol{\vartheta}_{t:T}])\middle|\mathcal{F}_t\right]\\
\text{(by Proposition 1)} &= \mathbb{E}\left[\vartheta_{t,i}\left(\Delta X_{t,i} + \frac{X_{t+1,i}}{\boldsymbol{\vartheta}_{t+1}^\top\boldsymbol{X}_{t+1}}\,c_{t+1}^{\boldsymbol{\theta}}\right.\right.\\
&\qquad\left.\left.\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]\right)\gamma_t(U_t[\boldsymbol{\vartheta}_{t:T}])\middle|\mathcal{F}_t\right]\\
&= c_t^{\boldsymbol{\theta}}\,\mathbb{E}\left[\theta_{t,i}\left(\Delta X_{t,i} + \frac{X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top\boldsymbol{X}_{t+1}}\right.\right.\\
&\qquad\left.\left.\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]\right)\gamma_t(U_t[\boldsymbol{\vartheta}_{t:T}])\middle|\mathcal{F}_t\right].
\end{aligned} \tag{10}$$

Next, $U_t[\boldsymbol{\vartheta}_{t:T}]$ may be simplified by noting that

$$\begin{aligned}\boldsymbol{\vartheta}_t^\top \Delta \boldsymbol{X}_t + w_t^{\boldsymbol{\vartheta}} \mathfrak{R}_{t+1}[\boldsymbol{\vartheta}_{t+1:T}] &= \boldsymbol{\vartheta}_t^\top \Delta \boldsymbol{X}_t + \mathfrak{R}_{t+1}[\boldsymbol{\vartheta}_{t+1:T}] \\ &\qquad \text{(as } \vartheta \text{ is self-financing)} \\ &\text{(by Proposition 1)} = \boldsymbol{\vartheta}_t^\top \Delta \boldsymbol{X}_t + c_{t+1}^{\boldsymbol{\theta}}\ \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}] \\ &\text{(as } \boldsymbol{\vartheta}_t = c_t^{\boldsymbol{\theta}}\ \boldsymbol{\theta}_t) = c_t^{\boldsymbol{\theta}}\ \boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + c_{t+1}^{\boldsymbol{\theta}}\ \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}] \\ &\text{(as } c_{t+1}^{\boldsymbol{\theta}} = c_t^{\boldsymbol{\theta}}\ w_t^{\boldsymbol{\theta}}) = c_t^{\boldsymbol{\theta}}(\boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + w_t^{\boldsymbol{\theta}} \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]).\end{aligned}$$

Furthermore, as $c_t^{\boldsymbol{\theta}} > 0$ a.s., the bivariate vector

$$(c_t(\boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + w_t^{\boldsymbol{\theta}} \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]),\ \boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + w_t^{\boldsymbol{\theta}} \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}])$$

is comonotonic,

and therefore, $U_t[\boldsymbol{\vartheta}_{t:T}] = U_t[\boldsymbol{\theta}_{t:T}]$ a.s.; that is, both $\boldsymbol{\theta}_{t:T}$ and $\boldsymbol{\vartheta}_{t:T}$ generate the same comonotonic uniform rv.

Finally, applying this result and continuing from (10), we have

$$\begin{aligned}&RC_{t,i}[\boldsymbol{\vartheta}] \\ &= c_t^{\boldsymbol{\theta}}\, \mathbb{E}\left[\theta_{t,i}\left(\Delta X_{t,i} + \frac{X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top \boldsymbol{X}_{t+1}} \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]\right)\gamma_t(U_t[\boldsymbol{\theta}_{t:T}])\middle|\mathcal{F}_t\right] \\ &= c_t^{\boldsymbol{\theta}}\, RC_{t,i}[\boldsymbol{\theta}_{t:T}]\end{aligned}$$

as required. □

Consequently, the risk contributions of the induced self-financing strategy are by definition positive homogeneous as stated in Proposition 3, and they satisfy the full allocation.

4. Dynamic Risk Budgeting Portfolios

Using the dynamic risk contributions defined in the last section, we now define a dynamic risk budgeting strategy.

Definition 7. Let $\{\rho_t\}_{t\in\mathcal{T}}$ be a DRM. A strategy $\boldsymbol{\theta}_{0:T} \in \mathcal{A}_c$, $c \geq 0$, is called a *dynamic risk budgeting strategy* with budget $B = (b_{t,i})_{t\in\mathcal{T}, i\in\mathcal{N}}$ satisfying $b_{t,i} > 0$ and $\sum_{i\in\mathcal{N}} b_{t,i} = 1$ for all $t \in \mathcal{T}$ if

$$RC_{t,i}[\boldsymbol{\theta}_{t:T}] = b_{t,i}\, \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}], \quad \forall t \in \mathcal{T}, \quad \text{and} \quad i \in \mathcal{N}. \tag{11}$$

A dynamic risk budgeting strategy is, therefore, a strategy such that at each time $t \in \mathcal{T}$, the risk contribution of investment $i \in \mathcal{N}$ is equal to $b_{t,i}$% of the risk-to-go at time t. For example, if the risk budget is $b_{t,i} = \frac{1}{n}$ for all $i \in \mathcal{N}$ and $t \in \mathcal{T}$, then we call the risk budgeting strategy *risk parity*, which means equal risk contribution because it satisfies

$$RC_{t,i}[\boldsymbol{\theta}_{t:T}] = RC_{t,j}[\boldsymbol{\theta}_{t:T}], \quad \forall i, j \in \mathcal{N}, \text{ and } \forall t \in \mathcal{T}.$$

Proposition 6. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be a coherent distortion DRM, $\boldsymbol{\theta}$ be an admissible strategy, and $\boldsymbol{\vartheta}$ be its corresponding induced self-financing strategy. If $\boldsymbol{\theta}$ is a risk budgeting strategy with risk budget B, then $\boldsymbol{\vartheta}$ is a risk budgeting strategy with risk budget B.*

Proof. The case when $t = 0$ is trivial because $\mathfrak{R}_0[\boldsymbol{\theta}_{0:T}] = \mathfrak{R}_0[\boldsymbol{\vartheta}_{0:T}]$ and $RC_{0,i}[\boldsymbol{\theta}_{0:T}] = RC_{0,i}[\boldsymbol{\vartheta}_{0:T}]$. Next, let $t > 0$, and assume that $\boldsymbol{\theta}_{0:T}$ is a risk budgeting strategy. Then, for each $t \in \mathcal{T}/\{0\}$ and $i \in \mathcal{N}$, it holds that

$$\begin{aligned}RC_{t,i}[\boldsymbol{\vartheta}_{t:T}] &= \left(\prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}}\right) RC_{t,i}[\boldsymbol{\theta}_{t:T}] = \left(\prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}}\right) b_{t,i}\, \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] \\ &= b_{t,i}\left(\prod_{s=0}^{t-1} w_s^{\boldsymbol{\theta}}\right) \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] = b_{t,i}\, \mathfrak{R}_t[\boldsymbol{\vartheta}_{t:T}],\end{aligned}$$

where we used Proposition 5 in the first equation, then the fact that $\boldsymbol{\theta}_{0:T}$ is a risk budgeting strategy, and finally, Proposition 1. □

The next result pertains to the characterization of self-financing risk budgeting strategies as a unique solution of a series of strictly convex and recursive (backward in time) optimization problems. Moreover, we show under technical conditions that if a self-financing risk budgeting strategy with initial wealth of one exists, then it is given by a rescaled version of the solution to the series of convex optimization problems.

Theorem 3. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be a coherent distortion DRM with weight functions $\{\gamma_t\}_{t\in\mathcal{T}}$ and $B = (b_{t,i})_{t\in\mathcal{T}, i\in\mathcal{N}}$ be a risk budget. For $c > 0$, consider the recursive optimization problems*

$$\boldsymbol{\theta}_t^* := \underset{\boldsymbol{\theta}_t \in \mathcal{A}_{c,t}}{\arg\min}\ \mathbb{E}\left[\mathfrak{R}_t[(\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T}^*)] - \sum_{i\in\mathcal{N}} b_{t,i} \log \theta_{t,i}\right], \quad \forall t \in \mathcal{T}. \tag{P}$$

There exists at most one solution to (P) for every $c > 0$. If there exists a solution to (P) with $\boldsymbol{\theta}^* \in \mathcal{A}_c^{\circ}$ (the interior of the set $\mathcal{A}_c \subset \mathcal{Z}_{0:T}$) for some $c > 0$, then it satisfies $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}^*] = 1$ a.s. for all $t \in \mathcal{T}$. Moreover,

a. *the self-financing strategy $\boldsymbol{\vartheta}_{0:T}^*$, induced by $\boldsymbol{\theta}_{0:T}^*$, is in $\mathcal{A}_0$ and is a self-financing risk budgeting strategy with risk budget B; and*

b. *the normalized strategy $\boldsymbol{\vartheta}_{0:T}^\dagger := \frac{1}{\boldsymbol{\vartheta}_0^\top \boldsymbol{X}_0} \boldsymbol{\vartheta}_{0:T}^*$ is in $\mathcal{A}_0$ and is a self-financing risk budgeting strategy with the same risk budget B and initial wealth of one.*

Proof. As the risk-to-go process is convex, recall that the one-step risk measures are convex, the $(-\log)$ is strictly convex, and the objective functional is strictly convex. Moreover, the set of admissible strategies is convex; thus, at most one solution to (P) exists.

For the remainder of the proof, we suppose that a solution to (P) with $\boldsymbol{\theta}^* \in \mathcal{A}_c^{\circ}$, for some $c > 0$, exists. The proof that the risk-to-go process is a.s. equal to one follows from the proof of part (a).

Part (a). For $t \in \mathcal{T}$, consider the objective function (where for ease of notation, we suppress the dependence of L_t on $\boldsymbol{\theta}_{t+1:T}^*$)

$$L_t[\boldsymbol{\theta}_t] := \mathbb{E}\left[\mathfrak{R}_t[(\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t+1:T}^*)] - \sum_{i\in\mathcal{N}} b_{t,i} \log \theta_{t,i}\right] \quad \text{for} \quad \boldsymbol{\theta}_t \in \mathcal{A}_{c,t}. \tag{12}$$

Next, take any $\boldsymbol{\theta}'_t \in \mathcal{A}_{c,t}$. Then, by Lemma A.2, it holds that for all $\varepsilon \in [0,1]$, the strategy $\boldsymbol{\theta}_t + \varepsilon(\boldsymbol{\theta}'_t - \boldsymbol{\theta}_t) = (1-\varepsilon)\boldsymbol{\theta}_t + \varepsilon\boldsymbol{\theta}'_t \in \mathcal{A}_{c,t}$. By Proposition A.2, we can interchange limit and expectation; thus, the one-sided Gâteaux derivative of $L_t[\boldsymbol{\theta}_t]$ in direction $\delta\boldsymbol{\theta} := (\boldsymbol{\theta}'_t - \boldsymbol{\theta}_t)$ is

$$\begin{aligned}
&\lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon}(L_t[\boldsymbol{\theta}_t + \varepsilon\,\delta\boldsymbol{\theta}] - L_t[\boldsymbol{\theta}_t]) \\
&= \sum_{i\in\mathcal{N}} \mathbb{E}\left[\mathcal{D}_i^{\delta\theta_i}\,\Re_t[(\boldsymbol{\theta}_t, \boldsymbol{\theta}^*_{t+1:T})] - b_{t,i}\,\frac{\delta\theta_i}{\theta_{t,i}}\right] \\
&= \sum_{i\in\mathcal{N}} \mathbb{E}\left[\mathbb{E}\left[\mathcal{D}_i^{\delta\theta_i}\,\Re_t[(\boldsymbol{\theta}_t, \boldsymbol{\theta}^*_{t+1:T})] - b_{t,i}\,\frac{\delta\theta_i}{\theta_{t,i}}\middle|\mathcal{F}_t\right]\right] \\
&= \sum_{i\in\mathcal{N}} \mathbb{E}\left[\delta\theta_i\,\mathbb{E}\left[\left(\Delta X_{t,i} + \frac{X_{t+1,i}}{\boldsymbol{\theta}^{*\top}_{t+1}\boldsymbol{X}_{t+1}}\Re_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\right)\right.\right. \\
&\qquad \left.\left.\gamma_t(U_t[(\boldsymbol{\theta}_t, \boldsymbol{\theta}^*_{t+1:T})]) - \frac{b_{t,i}}{\theta_{t,i}}\middle|\mathcal{F}_t\right]\right]. \qquad (13)
\end{aligned}$$

As $\mathcal{A}_{c,t}$ is a convex set and we assume that $\boldsymbol{\theta}^*_t \in \mathcal{A}^{\circ}_{c,t}$, the unique optima $\boldsymbol{\theta}^*_t$ is attained where the one-sided Gâteaux derivative is nonnegative for all $\boldsymbol{\theta}'_t \in \mathcal{A}_{c,t}$ (see, e.g., Rockafellar 2015, theorem 23.2): that is,

$$\lim_{\varepsilon\downarrow 0}\frac{1}{\varepsilon}(L_t[\boldsymbol{\theta}^*_t + \varepsilon(\boldsymbol{\theta}'_t - \boldsymbol{\theta}^*_t)] - L_t[\boldsymbol{\theta}^*_t]) \geq 0, \quad \forall \boldsymbol{\theta}'_t \in \mathcal{A}_{c,t},$$

which by Equation (13), is equivalent to

$$\sum_{i\in\mathcal{N}} \mathbb{E}\left[(\theta'_{t,i} - \theta^*_{t,i})\,\frac{RC_{t,i}[\boldsymbol{\theta}^*_{t:T}] - b_{t,i}}{\theta^*_{t,i}}\right] \geq 0, \quad \forall \boldsymbol{\theta}'_t \in \mathcal{A}_{c,t}. \qquad (14)$$

We next show that $\boldsymbol{\theta}^*_t$ fulfils (14) if and only if it satisfies

$$RC_{t,i}[\boldsymbol{\theta}^*_{t,T}] = b_{t,i}, \qquad \forall i \in \mathcal{N}. \qquad (15)$$

To prove this, note that if $\boldsymbol{\theta}^*_t$ satisfies (15), then (14) holds. To show that (14) implies (15), we proceed by contradiction. Suppose the equality in (15) does not hold a.s. for some $i = k$, and let $B^+ \in \mathcal{F}$, $B^- \in \mathcal{F}$, $B^0 \in \mathcal{F}$ denote the sets on which $(RC_{t,k}[\boldsymbol{\theta}^*_{t,T}] - b_{t,k})$ is positive, negative, and zero, respectively. By assumption, $\mathbb{P}(B^+) + \mathbb{P}(B^-) > 0$. Then, define $\boldsymbol{\theta}'_t$ s.t. $\theta'_{t,i} = \theta^*_{t,i}$ for all $i \neq k$ and

$$\theta'_{t,k} := 2\,\theta^*_{t,k}\mathbb{1}_{B^-} + \frac{1}{2}(c + \theta^*_{t,k})\,\mathbb{1}_{B^+} + \theta^*_{t,k}\mathbb{1}_{B^0}.$$

Note that $\boldsymbol{\theta}'_t \geq c$ a.s.; moreover, $\boldsymbol{\theta}'_t \leq 2\,\boldsymbol{\theta}^*_t$. Hence, we have that $\mathbb{E}[\sum_{i\in\mathcal{N}}(\theta'_{t,i}\Delta X_{t,i})^2] \leq 4\mathbb{E}[\sum_{i\in\mathcal{N}}(\theta^*_{t,i}\Delta X_{t,i})^2] < +\infty$ and $\frac{\boldsymbol{\theta}'^{\top}_t \boldsymbol{X}_{t+1}}{\boldsymbol{\theta}^{*\top}_{t+1}\boldsymbol{X}_{t+1}} \leq \frac{2\boldsymbol{\theta}^{*\top}_t \boldsymbol{X}_{t+1}}{\boldsymbol{\theta}^{*\top}_{t+1}\boldsymbol{X}_{t+1}} = 2\,w^{\boldsymbol{\theta}^*}_t < +\infty$. Therefore, $\boldsymbol{\theta}'_t \in \mathcal{A}_{c,t}$. However,

$$\begin{aligned}
&\lim_{\varepsilon\downarrow 0}\frac{1}{\varepsilon}(L_t[\boldsymbol{\theta}^*_t + \varepsilon(\boldsymbol{\theta}'_t - \boldsymbol{\theta}^*_t)] - L_t[\boldsymbol{\theta}^*_t]) \\
&= \mathbb{E}\left[(\theta^*_{t,k}\mathbb{1}_{B^-} + \frac{1}{2}(c - \theta^*_{t,k})\,\mathbb{1}_{B^+})\frac{(RC_{t,k}[\boldsymbol{\theta}^*_{t,T}] - b_{t,k})}{\theta^*_{t,k}}\right] < 0,
\end{aligned}$$

and therefore, $\boldsymbol{\theta}^*_t$ does not satisfy (14); we arrive at a contradiction.

Thus, $\boldsymbol{\theta}^*_t$ must satisfy Equation (15); by Corollary 1, it holds that

$$\Re_t[\boldsymbol{\theta}^*_{t:T}] = \sum_{i\in\mathcal{N}} RC_{t,i}[\boldsymbol{\theta}^*_{t:T}] = \sum_{i\in\mathcal{N}} b_{t,i} = 1,$$

$$\text{which implies that} \quad RC_{t,i}[\boldsymbol{\theta}^*_{t:T}] = b_{t,i}\,\Re_t[\boldsymbol{\theta}^*_{t:T}],$$

and $\boldsymbol{\theta}^*_{t:T}$ satisfies the risk budgeting Equation (11) for all $t \in \mathcal{T}$.

Finally, by Proposition 6, the self-financing strategy $\boldsymbol{\vartheta}^*_{0:T}$ induced by $\boldsymbol{\theta}^*_{0:T}$ is a risk budgeting strategy with budget B. Moreover, as $\boldsymbol{\theta}^*_{0:T} \in \mathcal{A}^{\circ}_c$, we have that $\boldsymbol{\vartheta}^*_{0:T} \in \mathcal{A}_0$.

Part (b). Define $\boldsymbol{\vartheta}^{\dagger}_{0:T} := \frac{1}{\boldsymbol{\vartheta}^{*\top}_0 \boldsymbol{X}_0}\boldsymbol{\vartheta}^*_{0:T}$; clearly, $\boldsymbol{\vartheta}^{\dagger}_{0:T}$ has initial wealth of one, is self-financing, and is an element of $\mathcal{A}_0$ as $\boldsymbol{\vartheta}^{*\top}_0 \boldsymbol{X}_0 > 0$, and we claim that it is a risk budgeting strategy with budget B. Indeed, we have for all $t \in \mathcal{T}$ and $i \in \mathcal{N}$,

$$\begin{aligned}
RC_{t,i}[\boldsymbol{\vartheta}^{\dagger}_{t:T}] &= \frac{RC_{t,i}[\boldsymbol{\vartheta}^*_{t:T}]}{\boldsymbol{\vartheta}^{*\top}_0 \boldsymbol{X}_0} = \frac{b_{t,i}}{\boldsymbol{\vartheta}^{*\top}_0 \boldsymbol{X}_0}\,\Re_t[\boldsymbol{\vartheta}^*_{t:T}] \\
&= b_{t,i}\left(\frac{1}{\boldsymbol{\vartheta}^{*\top}_0 \boldsymbol{X}_0}\Re_t[\boldsymbol{\vartheta}^*_{t:T}]\right) = b_{t,i}\,\Re_t[\boldsymbol{\vartheta}^{\dagger}_{t:T}],
\end{aligned}$$

where we applied positive homogeneity of the risk contributions (see Proposition 3) and the fact that $\boldsymbol{\vartheta}^*_{0:T}$ is a risk budgeting strategy. Thus, $\boldsymbol{\vartheta}^{\dagger}_{0:T}$ is in $\mathcal{A}_0$ and a self-financing risk budgeting strategy with risk budget B and initial wealth of one. □

The above result states that if an optimizer of (P) exists, it is a risk budgeting strategy.

Remark 1. If we add an extra constraint that $\theta_{t,i}$ are bounded above by a constant $d \in (c, +\infty)$ for all $t \in \mathcal{T}$, then we immediately have that $w^{\boldsymbol{\theta}}_t \leq \frac{d}{c} < +\infty$ for all $t \in \mathcal{T}$. That is, the weights are uniformly bounded. This results in the set of admissible strategies becoming closed, which ensures the existence of a unique solution to (P).

In the next theorem, whose proof is deferred to Appendix B.2, we show that under technical conditions, any self-financing risk budgeting strategy with initial wealth of one is a rescaled version to the solution of the optimization Problem (P) and in particular, given in Theorem 3(b).

Theorem 4 (Uniqueness). *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be a coherent distortion DRM. Consider an admissible self-financing dynamic risk budgeting strategy $\boldsymbol{\varphi}_{0:T} \in \mathcal{A}^{\circ}_c$ for some $c > 0$ that has initial wealth of one and budget B. If the corresponding risk-to-go process satisfies $0 < c_R \leq \Re_t[\boldsymbol{\varphi}_{t:T}] \leq c^R < +\infty$ for some $c_R, c^R \in (0, +\infty)$ and all $t \in \mathcal{T}$, then the risk budgeting strategy is the unique solution to (P) with the lower bound on the strategy $\frac{c}{c^R}$ and characterized by Theorem 3(b).*

5. Approximation of Risk Budgeting Strategies

Although Theorems 3 and 4 provide a characterization of risk budgeting strategies as solutions to a sequence of convex optimization problems, they do not provide a methodology for finding them. Thus, we develop a deep learning approach that leverages the flexibility of neural networks to approximate high-dimensional functions (see, e.g., Goodfellow et al. 2016) together with new techniques that have been developed for optimizing convex DRMs (Coache and Jaimungal 2023, Coache et al. 2023). The latter works solve portfolio allocation and algorithmic trading problems by making explicit use of the dual representation of convex risk measures. In contrast, our proposed approach relies on the analytical results developed for the class of coherent distortion DRM and in particular, their risk contributions given in Theorem 2.

To optimize the performance criterion $L_t[\boldsymbol{\theta}_{t:T}]$ (given in Equation (12)) of (P), we make use of gradient descent methods. The Gâteaux derivative is related to the risk contributions (see Theorem 2) and requires evaluating the risk-to-go $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]$ and the rv $U_t[\boldsymbol{\theta}_{t:T}]$. Therefore, we develop algorithms for estimating and sampling from $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]$ and $U_t[\boldsymbol{\theta}_{t:T}]$ and for estimating the Gâteaux derivative of $L_t[\boldsymbol{\theta}_{t:T}]$.

The general strategy behind our actor-critic algorithm is as follows.

1. Parameterize the strategy $\boldsymbol{\theta}_{0:T}$ by an NN with parameters $\boldsymbol{\beta}$.

2. For estimating $U_t[\boldsymbol{\theta}_{t:T}]$, we approximate the conditional cdf of $g_t := \boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + w_t^{\boldsymbol{\theta}} \mathfrak{R}_{t+1}$ given $\mathcal{F}_t$ (i.e., $F_{g_t|\mathcal{F}_t}(z) := \mathbb{P}(g_t \le z|\mathcal{F}_t)$) by an NN with parameters $\mathfrak{f}$.

3. For estimating $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]$, we approximate the risk-to-go by an NN with parameters $\mathfrak{r}$.

To implement points (2) and (3), we use the notion of elicitability (see Section 5.2), which provides a numerically efficient alternative to nested simulations when calculating conditional risk measures and cdfs (Coache et al. 2023). Specifically, we use Proposition C.4 for approximating $F_{g_t|\mathcal{F}_t}(\cdot)$ and Proposition C.3 for approximating $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]$.

The parameterized strategies $\boldsymbol{\theta}_{0:T}$ are the so-called "actors" as they are at the investor's discretion. The approximation of the conditional cdf and the risk-to-go are the "critics" as they evaluate the effectiveness of a strategy. In the following subsections, we provide details on the algorithm and the gradients for training the NNs (Section 5.1), how conditional elicitability is leveraged in the algorithm (Section 5.2), and the specific NN architectures used (Section 5.3).

5.1. Algorithms and Gradient Formulas

Here, we explain the general structure of the algorithm. Optimization of the criterion $L_t[\boldsymbol{\theta}_{t:T}]$ proceeds in an iterative fashion, where for each iteration, we update (i) the risk-to-go m_r times, then (ii) the conditional cdf m_f times, and finally, (iii) the strategy once. For the risk-to-go, we employ a main network and a target network (with parameters $\mathfrak{r}$ and $\mathfrak{r}'$, respectively), which are known to improve stability and convergence rates of actor-critic methods (Mnih et al. 2015). The main network of the risk-to-go is updated using stochastic gradient descent, whereas the target network is updated using soft updates (i.e., via a convex combination of the current main and target network parameters, $\mathfrak{r}' \leftarrow (1-\tau)\mathfrak{r}' + \tau\,\mathfrak{r}$, with soft-update rate $\tau \in (0,1)$ (Fujita et al. 2021)). The detailed algorithm is provided in Algorithms 1 and 2.

Algorithm 1 (Actor-Critic Algorithm for Learning Risk Budgeting Strategies)

Input: NN parameters $\boldsymbol{\beta}$, $\mathfrak{r}$, $\mathfrak{r}'$, and $\mathfrak{f}$; number of risk-to-go updates $m_r \ge 1$ per iteration; number of conditional cdf updates $m_f \ge 1$ per iteration; learning rates; soft-update rate τ

do
  for $i = 1, 2, \ldots, m_r$ **do**
    get simulations of $\boldsymbol{X}_{0:T}$, $\boldsymbol{\Theta}_{0:T}$, and $(\mathfrak{R}_t)_{t\in\mathcal{T}}$ using Algorithm 2;
    compute the expected score S_ρ (18) using the risk-to-go with main parameters $\mathfrak{r}$;
    use a gradient step to minimize the score and update the risk-to-go main parameters $\mathfrak{r}$;
    perform a soft update of the target to the main NN parameters $\mathfrak{r}' \leftarrow (1-\tau)\,\mathfrak{r}' + \tau\,\mathfrak{r}$;
  for $j = 1, 2, \ldots, m_f$ **do**
    get simulations of $\boldsymbol{X}_{0:T}$, $\boldsymbol{\Theta}_{0:T}$, and $(\mathfrak{R}_t)_{t\in\mathcal{T}}$ using Algorithm 2;
    compute the expected score S_{cdf} (19) using conditional cdf parameters $\mathfrak{f}$;
    use a gradient step to minimize the score and update the cdf parameters $\mathfrak{f}$;
  get simulations of $\boldsymbol{X}_{0:T}$, $\boldsymbol{\Theta}_{0:T}$, and $(\mathfrak{R}_t)_{t\in\mathcal{T}}$ using Algorithm 2;
  perform a gradient step to minimize the objective (12) using (16) and update the strategy parameters $\boldsymbol{\beta}$;
while *not converged*;

Algorithm 2 (Simulate Asset Prices, Strategy, and Risk-to-Go)

Input: NN parameters $\boldsymbol{\beta}$ and $\mathfrak{r}'$;
simulate asset prices $\boldsymbol{X}_{0:T}$;
use asset prices to generate samples of the strategy $\boldsymbol{\Theta}_{0:T}$ with policy parameters $\boldsymbol{\beta}$;
compute the risk-to-go for all simulations and time points with NN target parameters $\mathfrak{r}'$;
Output: asset prices: $\boldsymbol{X}_{0:T}$, strategies: $\boldsymbol{\Theta}_{0:T}$, risk-to-go: $(\mathfrak{R}_t)_{t\in\mathcal{T}}$

Next, we discuss how the NN parameters of the strategy $\boldsymbol{\theta}_{0:T}$ are trained. Recall that the strategy's parameters are a set of vectors $(\boldsymbol{\beta}_t)_{t\in\mathcal{T}}$, where for all $t \in \mathcal{T}$,

we have $\boldsymbol{\beta}_t \in \mathcal{B}$, $\mathcal{B} \subset \mathbb{R}^m$. We write, with slight abuse of notation, $\boldsymbol{\theta}_t^{\boldsymbol{\beta}_t} = \boldsymbol{\theta}_t(X_0, \dots, X_t; \boldsymbol{\beta}_t)$, where $\boldsymbol{\theta}_t : \mathbb{R}^{n\times t} \times \mathcal{B} \to \mathbb{R}$. We call $\boldsymbol{\beta}_{0:T}$ a policy as it parametrizes the investor's strategy. Next, for $t \in \mathcal{T}$, we view the criterion in Equation (12) as a function of $\boldsymbol{\beta}_t$ and aim to minimize it over these parameters. To this end, we write the time t loss function as

$$L_t[\boldsymbol{\beta}_t; \boldsymbol{\beta}^*_{t+1:T}] := \mathbb{E}\left[\Re_t[(\boldsymbol{\theta}_t^{\boldsymbol{\beta}_t}, \boldsymbol{\theta}_{t+1}^{\boldsymbol{\beta}^*_{t+1}}, \dots, \boldsymbol{\theta}_T^{\boldsymbol{\beta}^*_T})] - \sum_{i\in\mathcal{N}} b_{t,i} \log \theta_{t,i}^{\boldsymbol{\beta}_t}\right],$$

where $\boldsymbol{\beta}^*_{t+1:T}$ are the optimal policies from time $t+1$ onward. We adopt for the methodology of deterministic policy gradient, and starting from the current estimate of the parameters, we update the parameters using the gradient step rule

$$\boldsymbol{\beta}_t \leftarrow \boldsymbol{\beta}_t - \eta \nabla_{\boldsymbol{\beta}} L_t[\boldsymbol{\beta}; \boldsymbol{\beta}^*_{t+1:T}]|_{\boldsymbol{\beta}=\boldsymbol{\beta}_t},$$

where $0 < \eta \ll 1$ is a learning rate. Hence, we require an efficient way to estimate the gradient of L_t. By the chain rule (as in the proof of Theorem 2), we have

$$\begin{aligned}
&\partial_{\beta_k} L_t[\boldsymbol{\beta}; \boldsymbol{\beta}^*_{t+1:T}] \\
&= \mathbb{E}\left[\partial_{\beta_k}\theta_{t,i}^{\boldsymbol{\beta}}\left\{\left(\Delta X_{t,i} + \frac{X_{t+1,i}}{(\boldsymbol{\theta}_{t+1}^{\boldsymbol{\beta}^*_{t+1}})^\top \boldsymbol{X}_{t+1}} \Re_{t+1}[(\boldsymbol{\theta}_{t+1}^{\boldsymbol{\beta}^*_{t+1}}, \dots, \boldsymbol{\theta}_T^{\boldsymbol{\beta}^*_T})]\right)\right.\right. \\
&\qquad \left.\left. \times \gamma_t(U_t[(\boldsymbol{\theta}_t^{\boldsymbol{\beta}}, \boldsymbol{\theta}_t^{\boldsymbol{\beta}^*_{t+1}}, \dots, \boldsymbol{\theta}_T^{\boldsymbol{\beta}^*_T})]) - \frac{b_{t,i}}{\theta_{t,i}^{\boldsymbol{\beta}}}\right\}\right]. \qquad (16)
\end{aligned}$$

Given a policy $\boldsymbol{\beta}$, we estimate the expectation in (16) using its sample average, which however, requires samples of $\Re_{t+1}[(\boldsymbol{\theta}_t^{\boldsymbol{\beta}^*_{t+1}} \dots, \boldsymbol{\theta}_T^{\boldsymbol{\beta}^*_T})]$ and $U_t[(\boldsymbol{\theta}_t^{\boldsymbol{\beta}}, \boldsymbol{\theta}_{t+1}^{\boldsymbol{\beta}^*_{t+1}} \dots, \boldsymbol{\theta}_T^{\boldsymbol{\beta}^*_T})]$. Thus, we next elaborate on how we leverage the notion of conditional elicitability to simulate from these rvs.

5.2. Conditional Elicitability

As the risk-to-go $\Re_t[\boldsymbol{\theta}_{t:T}] = \rho_t(\boldsymbol{\theta}_t^\top \Delta \boldsymbol{X}_t + w_t^{\boldsymbol{\theta}} \Re_{t+1}[\boldsymbol{\theta}_{t+1:T}])$ consists of one-step (conditional) risk measures, both the risk-to-go and the conditional cdf could be estimated via nested simulations, which however, is numerically expensive. To overcome this, we use the notion of elicitable functionals, which circumvents the need for nested simulations. The key concept is that a functional is elicitable (e.g., mean, VaR_α, ES_α) if it is the minimizer of the expected value of a scoring function, where the Value-at-Risk of a rv Y is given by $\mathrm{VaR}_\alpha(Y) = F_Y^{-1}(\alpha)$. Thus, an elicitable functional may be estimated by solving a convex optimization problem. Furthermore, conditional elicitable functionals can be estimated by minimizing the expected score over arbitrary functions of the conditioning variables (see, e.g., Proposition 7). For completeness, in Appendix C, we collect known results on elicitability and conditional elicitability that are relevant to the exposition as well as new results pertaining to the specific risk measures considered here.

A large class of one-step distortion risk measures is elicitable—in particular, those with piecewise constant weight function (see Fissler and Ziegel 2016 for the static setting and Coache et al. 2023 for the dynamic setting). In the numerical examples, we consider a subclass of coherent distortion DRM given by the weighted average of the ES and expected value, but our approach can be generalized to other elicitable coherent distortion DRMs. Specifically, we consider the family of coherent distortion DRMs $\{\rho_t\}_{t\in\mathcal{T}}$ parametrized by $p \in [0,1]$:

$$\rho_t(Z) := p\,\mathrm{ES}_\alpha(Z|\mathcal{F}_t) + (1-p)\,\mathbb{E}[Z|\mathcal{F}_t], \quad Z \in \mathcal{Z}_{t+1}, \qquad (17)$$

which for each $t \in \mathcal{T}$, is a one-step distortion risk measure with weight function $\gamma_t(u) = p\,\frac{1}{1-\alpha}\mathbb{1}_{u\geq\alpha} + (1-p)$. For each $p \in [0,1]$, $\rho_t(\cdot)$ is coherent because the distortion weight function $\gamma_t(\cdot)$ is increasing. If $p = 0$, then $\rho_t(\cdot) = \mathbb{E}[\cdot|\mathcal{F}_t]$, and if $p = 1$, then $\rho_t(\cdot) = \mathrm{ES}_\alpha(\cdot|\mathcal{F}_t)$.

For the remainder of this subsection, let $(\boldsymbol{X}, Y)$ be a $\mathcal{F}$-measurable random vector with joint cdf $F_{\boldsymbol{X},Y}$, where Y is a univariate rv with cdf F_Y and $\boldsymbol{X}$ is an n-dimensional random vector with cdf $F_{\boldsymbol{X}}$. The next result explains how the one-step risk measures in (17) can be estimated using conditional elicitability.

Proposition 7 (Mean ES Risk Measure). *Let ρ_t be given in (17) with $p \in (0,1)$. Denote by $\mathcal{G} := \{g | g : \mathbb{R}^n \to \mathbb{R}\}$. Then, it holds that*

$$\begin{aligned}
&(\mathrm{VaR}_\alpha(Y|\boldsymbol{X}), \mathrm{ES}_\alpha(Y|\boldsymbol{X}), \rho_t(Y|\boldsymbol{X})) \\
&\quad = \operatorname*{arg\,min}_{(g_1,g_2,g_3)\in\mathcal{G}\times\mathcal{G}\times\mathcal{G}} \mathbb{E}[S_\rho(g_1(\boldsymbol{X}), g_2(\boldsymbol{X}), g_3(\boldsymbol{X}), Y)],
\end{aligned}$$

where for $0 < D < +\infty$, S_ρ is the strictly consistent scoring function given by

$$\begin{aligned}
S_\rho(z_1, z_2, z_3, y) := &\log\left(\frac{z_2 + D}{y + D}\right) - \frac{z_2}{z_2 + D} \\
&+ \frac{(\mathbb{1}_{\{y\leq z_1\}} - \alpha)z_1 + \mathbb{1}_{\{y>z_1\}}\, y}{(z_2 + D)(1-\alpha)} \\
&+ \left(\frac{z_3 - p\, z_2}{1-p} - y\right)^2. \qquad (18)
\end{aligned}$$

Thus, to estimate $\rho_t(Y|\boldsymbol{X})$, we proceed similarly to Coache et al. (2023) and Fissler and Pesenti (2023) and parameterize each function g_i with an NN $\mathfrak{g}^i$ with parameters $\boldsymbol{\nu}_i$, $i = 1,2,3$. Then, the NN parameters are estimated via

$$\begin{aligned}
(\hat{\boldsymbol{\nu}}_1, \hat{\boldsymbol{\nu}}_2, \hat{\boldsymbol{\nu}}_3) = \operatorname*{arg\,min}_{(\boldsymbol{\nu}_1,\boldsymbol{\nu}_2,\boldsymbol{\nu}_3)} \frac{1}{N}\sum_{k=1}^N S_\rho(&\mathfrak{g}^1_{\boldsymbol{\nu}_1}(\boldsymbol{x}^{(k)}), \mathfrak{g}^2_{\boldsymbol{\nu}_2}(\boldsymbol{x}^{(k)}), \\
&\mathfrak{g}^3_{\boldsymbol{\nu}_3}(\boldsymbol{x}^{(k)}), y^{(k)})
\end{aligned}$$

over independent simulated mini batches $(\boldsymbol{x}^{(k)}, y^{(k)})_{k\in\{1,\dots,N\}}$ of $(\boldsymbol{X}, Y)$. As ES_α is always larger than VaR_α, we set $\mathfrak{g}^2_{\boldsymbol{\nu}_2} := \mathfrak{g}^1_{\boldsymbol{\nu}_1} + \mathfrak{g}^4_{\boldsymbol{\nu}_2}$, where $\mathfrak{g}^4_{\boldsymbol{\nu}_2}$ is an NN with parameters $\boldsymbol{\nu}_2$ and a soft-plus output layer so that $\mathfrak{g}^4_{\boldsymbol{\nu}_2}(\cdot) \geq 0$.

Elicitability of cdfs was shown in Gneiting and Raftery (2007); here, we provide its conditional version (see also Proposition C.4 in Appendix C).

Proposition 8 (Conditional Distribution Function). *Denote by* $\mathcal{H} := \{h | h : \mathbb{R}^n \times \mathbb{R} \to [0,1], h(\cdot, z)$ *increasing in* $z\}$. *Then, it holds that*

$$F_{Y|\boldsymbol{X}}(\cdot) = \arg\min_{F \in \mathcal{H}} \mathbb{E}[S_{cdf}(F(\boldsymbol{X}, \cdot), Y)], \quad \text{where}$$
$$S_{cdf}(\boldsymbol{x}, y) := \int_{\mathbb{R}} (F(\boldsymbol{x}, z) - \mathbb{1}_{z \geq y})^2 \, dz. \tag{19}$$

As above, we parameterize the functions $F \in \mathcal{H}$ by NNs $\mathfrak{F}$ with parameters $\mathfrak{f}$ and estimate the parameters by

$$\hat{\mathfrak{f}} = \arg\min_{\mathfrak{f}} \frac{1}{N} \sum_{k=1}^{N} \sum_{l=1}^{L} (\mathfrak{F}_{\mathfrak{f}}(\boldsymbol{x}^{(k)}, z_l) - \mathbb{1}_{z_l \geq y^{(k)}})^2 \Delta z$$

over independent simulated mini batches $(\boldsymbol{x}^{(k)}, y^{(k)})_{k \in \{1,\dots,N\}}$, of $(\boldsymbol{X}, Y)$ and where $z_l := \underline{z} + (l-1)\Delta z$, $l = 1, \dots, L$ with $\Delta z := \frac{1}{L-1}(\overline{z} - \underline{z})$ for truncation limits $\underline{z}, \overline{z}$ such that $\underline{z} < \overline{z}$. The inner summation is an approximation to the Riemann integral in (19), whereas the outer summation is an empirical approximation to the expectation.

5.3. Neural Network Approximators

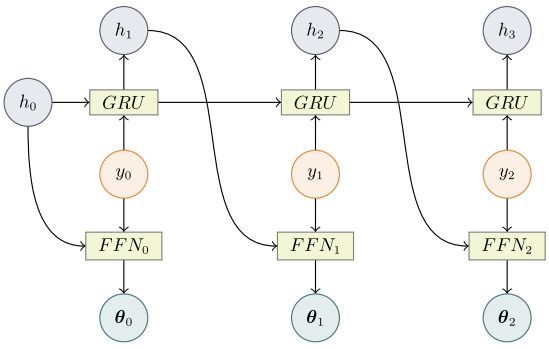
This section focuses on NN architectures for the strategy $\boldsymbol{\theta}_{0:T}$, the risk-to-go process $\mathfrak{R}_{0:T}$, and the uniform rvs $(U_t[\boldsymbol{\theta}_{t:T}])_{t \in \mathcal{T}}$. As it is not clear whether the optimal strategy or the risk-to-go process is Markovian in asset prices, we proceed using non-Markovian parameterizations. In the context of NN approximations, recurrent NNs can be used to accomplish this goal. Our implementation employs gated recurrent units (GRUs) to encode non-Markovian features through long short-term memory networks (attention networks are viable alternatives). Below, we describe the architecture for the actor-critic approach in detail.

First, the actor (strategy) NN (visualized in Figure 1) consists of a five-layered GRUs, with each layer consisting of hidden states of dimension n (recall that n is the asset dimension). The input features in the GRU are time, the wealth process of the induced self-financing strategy, and asset prices. We denote them by $y_t = (t, \ \boldsymbol{\vartheta}_{t-1}^{\top} \boldsymbol{X}_t, \boldsymbol{X}_t) \in \mathbb{R}^{n+2}$, $t \in \mathcal{T}$, and we call them the state. At each time $t \in \mathcal{T}$, the output from all hidden layers from the previous time step, denoted by h_{t-1}, and the state from the current time step, y_t, are concatenated and passed through a five-layer feed-forward neural network (FFN) to produce an n-dimensional output corresponding to $\boldsymbol{\theta}_t$. The internal layers of the FFN have sigmoid linear unit activation functions, whereas to ensure that the strategy is long only, the last layer has a soft-plus activation function.

Next, the critic (risk-to-go) NN (visualized in Figure 2) has the same GRU and FFN structure as the strategy network; however, the final output of the FFN is three dimensional, corresponding to the conditional VaR (VaR_t in Figure 2 and $\mathfrak{g}^1$ in Proposition 7), the difference between the conditional ES and the conditional VaR (ES_t in Figure 2 and $\mathfrak{g}^4$ in Proposition 7), and the conditional risk measure ($\mathfrak{R}_t$ in Figure 2 and $\mathfrak{g}^3$ in Proposition 7). There is no activation function in the final layer for VaR and the risk measure, whereas we have a soft-plus activation for the difference of ES and VaR to ensure that ES is always larger than or equal to VaR.

The second critic's NN architecture (for the conditional cdf $F_{g_t|\mathcal{F}_t}(\cdot)$) is provided in Figure 3 and is similar to that of the risk-to-go. Two important differences are that (i) we concatenate not only the hidden layers from the previous time step and the state but also, the value z corresponding to $F_{g_t|\mathcal{F}_t}(z)$ and that (ii) the output activation function is a sigmoid to ensure that $F_{g_t|\mathcal{F}_t}(z) \in (0,1)$. We also add the additional penalty $\int \mathbb{1}_{\{\partial_z F(\boldsymbol{x},z) < 0\}} (\partial_z F(\boldsymbol{x}, z))^2 \, dz$ to the score in Proposition C.4 to ensure that $F_{g_t|\mathcal{F}_t}(z)$ is increasing in z.

Figure 1. (Color online) Directed Graph Representation for Encodings and Parameterization of $\boldsymbol{\theta}_{0:T}$ Functions



Note. FFN, feed-forward neural network; GRU, gated recurrent unit.

6. Numerical Illustrations

In this section, we explore a stochastic volatility market model, an investor searching for a risk parity allocation (i.e., $b_{t,i} = \frac{1}{n}$ for all t and i), and a risk budgeting strategy

Figure 2. (Color online) Directed Graph Representation for Encodings and Parameterization of the Risk-to-Go $\mathfrak{R}_t$, Conditional ES, and Conditional VaR

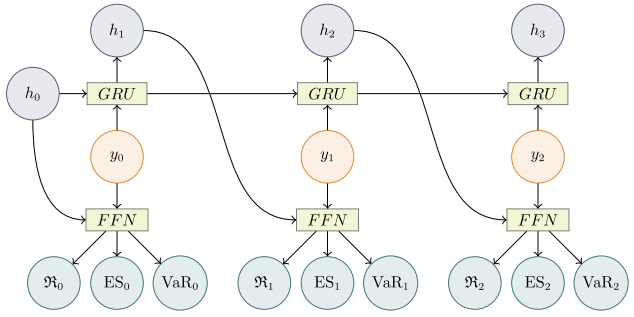
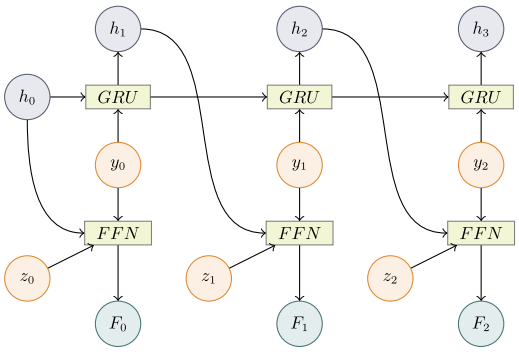


Figure 3. (Color online) Directed Graph Representation for Encodings and Parameterization of $F_t(\cdot) := F_{g_t|\mathcal{F}_t}(\cdot)$



with budget $\boldsymbol{b}_t = \left(\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{4}{15}, \frac{5}{15}\right)$ under the coherent distortion DRM given in (17). We consider $n = 5$ assets and a time horizon of $T + 1 = 12$, which corresponds to one year and monthly time steps. First, we describe the market model and then, the optimal risk budgeting strategy.

6.1. Market Model and NN Hyperparameters

We use a discrete time version of a Heston-inspired market model over a time horizon of $T + 1 = 12$ months, where asset returns have a Student t copula dependence. The investor updates their strategy monthly corresponding to time index $t \in \mathcal{T}$ in our optimization problem. For simulating the market model, we use the real-time variable $t_k := k\Delta t$ (with $\Delta t = \frac{1}{48}$ year) such that $t \in \mathcal{T}$ corresponds to a real time point of t_{4t} (i.e., we take four discretization steps between decision points). Specifically, we use a model inspired by the Milstein discretization of the Heston model and assume that

$$\log\left(\frac{X_{t_{k+1},i}}{X_{t_k,i}}\right) = (\mu_i - 0.5(v_{t_k,i})_+^2)\Delta t + \sqrt{(v_{t_k,i})_+}\,\Delta W^X_{t_k,i},$$

$$v_{t_{k+1},i} = \theta_i + ((v_{t_k,i})_+ - \theta_i)\,e^{-\kappa_i \Delta t} + \eta_i \sqrt{(v_{t_k,i})_+}\,\Delta W^v_{t_k,i} + \frac{1}{4}\eta_i^2((\Delta W^v_{t_k,i})^2 - \Delta t).$$

Here, $(\cdot)_+ := \max(\cdot, 0)$, $(\Delta W^X_{t_k,i}, \Delta W^v_{t_k,i})_{i\in\mathcal{N}}$ are independent across k but not i rvs. They are marginally normal with mean of zero and variance Δt. For $i \neq j$ and $t \in \mathcal{T}$, we have that (a) $\Delta W^X_{t_k,i}$ and $\Delta W^v_{t_k,j}$ are independent and that (b) $\Delta W^v_{t_k,i}$ and $\Delta W^v_{t_k,j}$ are independent. Moreover, $(\Delta W^X_{t_k,i}, \Delta W^X_{t_k,j})_{i,j\in\mathcal{N}}$ have a Student t copula with four degrees of freedom, and $(\Delta W^X_{t_k,i}, \Delta W^v_{t_k,i})_{i\in\mathcal{N}}$ follow a Gaussian copula. The corresponding correlation matrix for the dependence structure and the additional market model parameters are provided in Appendix D.1.

Figure 4. (Color online) Distribution of Log Returns of Various Assets

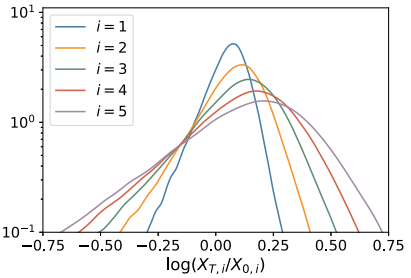


Figure 4 shows the distribution of the terminal log return, whereas Table 1 provides basic statistics of the asset's total return. As can be seen in Figure 4, the distributions are all left skewed, and volatility and expected return increase with the asset index label i.

When training the NNs, we use a learning rate of 0.001, a soft-update parameter of $\tau = 0.001$, the Adaptive Moment Estimation Weighted method for computing gradient updates of the NN parameters, and a scheduler that decreases the learning rate by a multiplicative factor of 0.99 every 20 outer iterations of Algorithm 1. As well, use $m_r = 20$ iterations for updating the risk-to-go and $m_f = 5$ iterations for updating the conditional cdfs. The specific NN architectures that we employ are as follows. (i) The GRUs have five layers, with each layer having five hidden states, and (ii) the feed-forward layers all have five layers, with 32 hidden nodes in each layer. We refer to Appendix D.2 for computational time metrics.

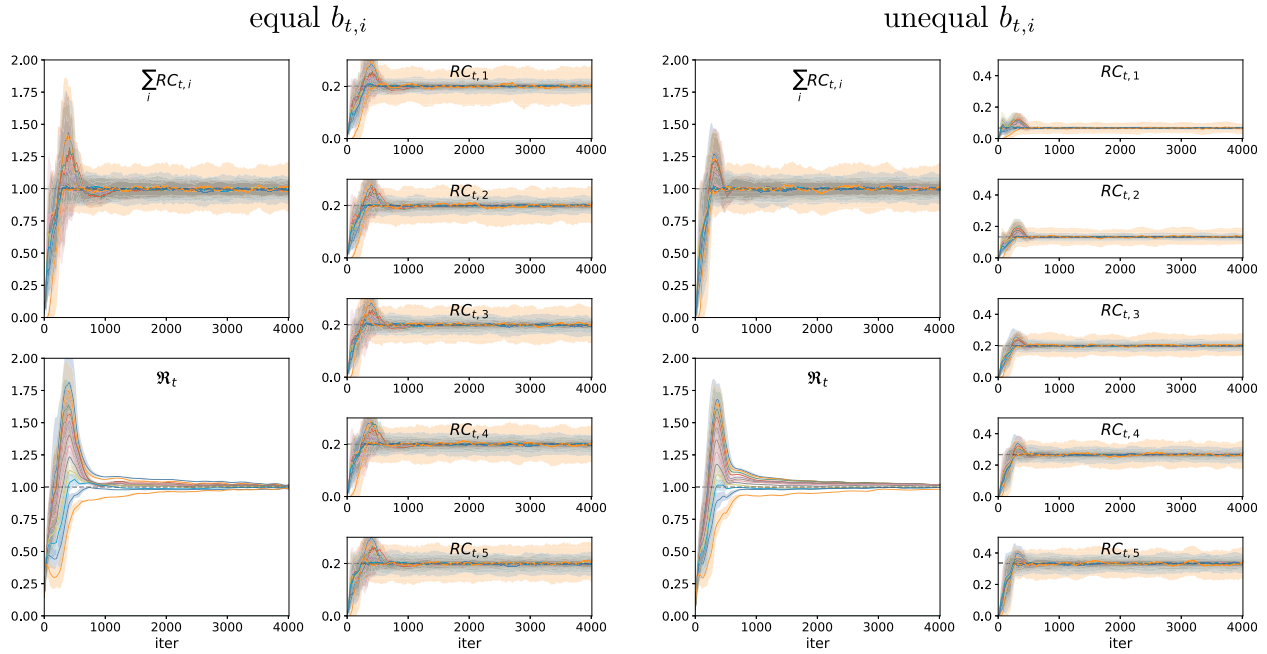
6.2. Risk Budgeting Strategy

In Figure 5, we provide convergence results for the case $p = 0.5$ and $\alpha = 0.75$. The x axes in Figure 5 are iterations, and for each $t \in \mathcal{T}$, we plot the risk contributions for all assets, the sum of risk contributions across assets, and the risk-to-go. The left panel of Figure 5 contains the results for the case when $\boldsymbol{b}_t = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right)$, and the right panel of Figure 5 contains the results for the case when $\boldsymbol{b}_t = \left(\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{4}{15}, \frac{5}{15}\right)$. The shaded

Table 1. Statistics of the Asset's Total Return $\frac{X_{T,i}}{X_{0,i}} - 1$

Asset	Mean	Standard deviation	Sharpe ratio	Correlation				
				$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$
$i = 1$	0.05	0.10	0.49	1				
$i = 2$	0.08	0.16	0.48	0.17	1			
$i = 3$	0.11	0.22	0.47	0.16	0.15	1		
$i = 4$	0.13	0.29	0.46	0.16	0.15	0.14	1	
$i = 5$	0.16	0.35	0.46	0.16	0.15	0.15	0.15	1

Figure 5. (Color online) Risk Contributions and Risk-to-Go vs. Iterations for the Optimal Strategy $\boldsymbol{\theta}^*_{0:11}$ When $p = 50\%$ and $\alpha = 75\%$



Notes. (Left panel) Equal risk contributions. (Right panel) $\boldsymbol{b}_t = \left(\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{4}{15}, \frac{5}{15}\right)$. $RC_{t,i}[\boldsymbol{\theta}_{t:T}]$ are estimated using 500 simulations at each iteration. The bands and lines correspond to the moving standard deviation and moving average (200 lags) of the corresponding quantities. iter, iteration.

regions in Figure 5 show a measure of the confidence in the estimator (that is, the standard deviation of the last 200 estimates resulting from the learned NN approximation of the risk-to-go and the simulated values of the risk contributions), whereas the solid lines in Figure 5 show the moving average using the last 200 estimates. As Figure 5 shows, the risk-to-go values all converge to the value of one, a result of Theorem 3, and the risk contributions all converge to $b_{t,i}$, a result of Equation (15). As the risk-to-go values all converge to the theoretical value of one and the risk contributions converge to their targeted values, this brings confidence that the numerical scheme has converged to a good approximation to the true solution of the problem. Interestingly, the risk contributions converge faster to their target values than the risk-to-go.

To gain a deeper understanding of the learned risk budgeting strategy, we present histograms in Figure 6 showing the percentage held in each asset across the 12 time steps. Each column in Figure 6 represents a fixed choice of p and a choice of the risk budget $b_{t,i}$, whereas the rows correspond to different assets.

In the first column of Figure 6, which corresponds to the case $p = 0.5$ and $b_{t,i} = \frac{1}{5}$, we observe a general trend, whereby the investment in asset i decreases as i increases. This trend is consistent with the fact that assets become increasingly volatile as the index i increases, making it reasonable to allocate less capital to the riskier assets to generate an equal risk budgeting portfolio. For the less risky assets $i = 1, 2, 3$, as time increases, investments become more disperse. Contrastingly, for the more risky assets $i = 4, 5$, as time increases, investments become more concentrated. This is sensible as the investor aims to have a risk parity portfolio at all points in time and hence, needs to deleverage the more risky assets. It is more challenging to provide a full description of how the distributions vary with p as there are a number of competing factors that are difficult to disentangle. If we fix, for example, the fifth row in Figure 6, $i = 5$, and compare the $p = 50\%$ case with the $p = 90\%$ case, where the investor puts more weight on the ES, the investment becomes less variable and more left skewed, meaning that they invest less in the most risky asset. If we fix, for example, the fourth row in Figure 6, $i = 4$, we observe that as p increases, the distribution of the percentage of wealth at time 2 becomes more variable but shifts to the left, once again indicating a deleverage. Focusing on the unequal $b_{t,i}$ cases in Figure 6, we see that the percentage of wealth invested in asset i increases as i increases, which is in line with the choices of $b_{t,i}$; they are now willing to take on more risk in the assets with a higher index.

Figure 7 provides an alternative view of the evolution of the weights invested in each asset. It shows the median percentage of wealth invested for all assets as a function of time together with the 20%–80% quantile bands for the same cases shown in Figure 6. Figure 7 shows that increasing p generally induces less variability in the percentage of wealth invested. Moreover,

Figure 6. (Color online) Percentage of Wealth Invested in Each Asset for Each Point in Time for $\alpha = 0.75$, $p = 50\%$ and 90%, $\boldsymbol{b}_t = (\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5})$, and $\boldsymbol{b}_t = (\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{4}{15}, \frac{5}{15})$

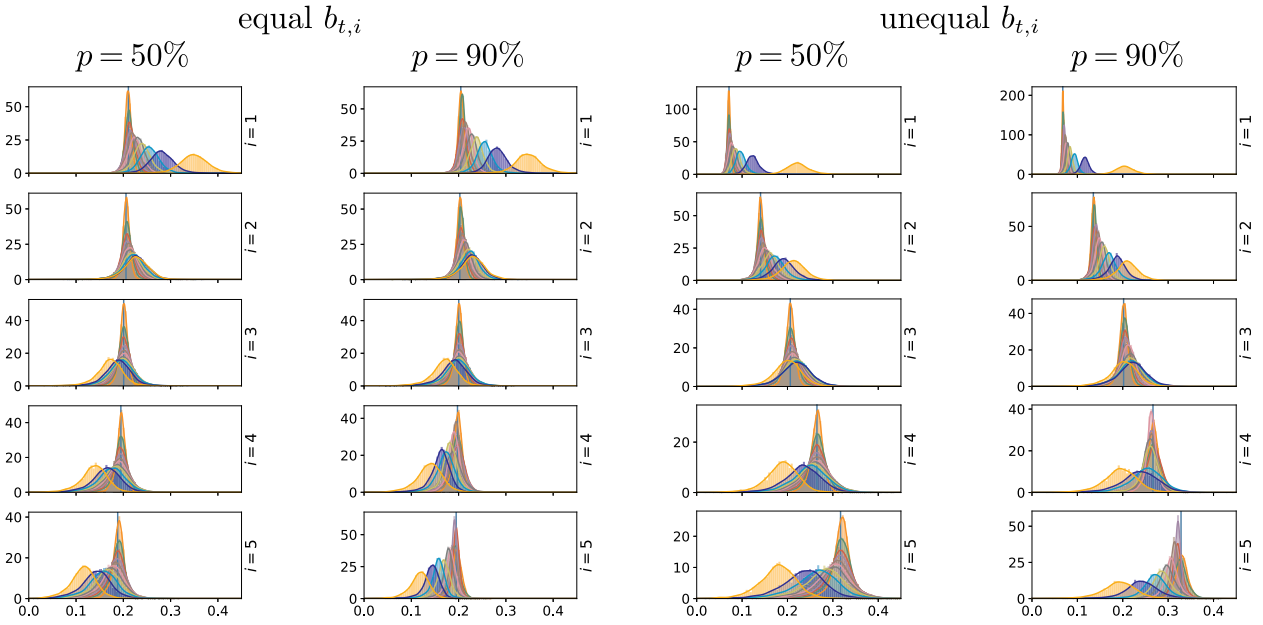
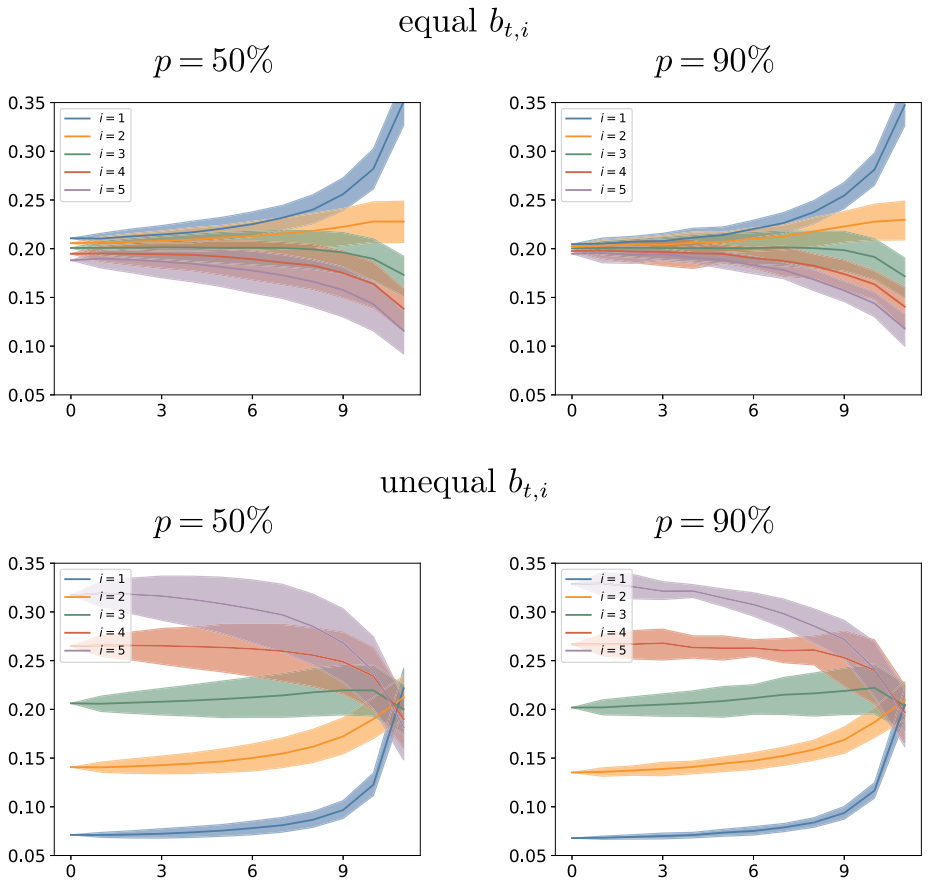


Figure 7. (Color online) Percentage of Wealth Invested in Each Asset for Each Point in Time for $\alpha = 0.75$, $p = 50\%$ and 90%, $\boldsymbol{b}_t = (\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5})$, and $(\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{4}{15}, \frac{5}{15})$



Note. The lines show the medians, and the bands show the 20% and 80% quantiles.

when the risk contributions are equal, the amounts invested start almost equally weighted but then spread out, whereas when the risk contributions are unequal, the amounts invested start unequal but move toward an equally weighted portfolio.

7. Conclusion

In this work, we show how an investor can allocate investments in risky assets to attain a predefined risk budget in a dynamic setting. To do so, we first propose a notion of risk contributions for coherent distortion DRM and demonstrate that they satisfy the full allocation property. For the class of coherent distortion DRM, we derive explicit formulas for risk contributions and prove that strategies that attain a particular risk budget are specified by the solution to a collection of convex optimization problems. Leveraging elicitability of coherent distortion DRM, we further provide a deep learning approach for solving those optimization problems. Finally, we demonstrate the stability of the numerical scheme through several examples using a stochastic volatility market model.

Appendix A. Auxiliary Definition and Results

The following definition of one-step distortion risk measures is in terms of the Choquet integral.

Definition A.1 (One-Step Distortion Risk Measures). For each $t \in \mathcal{T}$, let $g_t : [0,1] \times \Omega \to [0,1]$ be a (state-dependent) distortion function such that for all $\omega \in \Omega$, the function $g_t(\cdot, \omega)$ is nondecreasing and satisfies $g_t(0,\omega) = 0$ and $g_t(1,\omega) = 1$. Further, we assume that the rv $g_t(x,\cdot) : \Omega \to [0,1]$ is $\mathcal{F}_t$ measurable for every $x \in [0,1]$ and for all $t \in \mathcal{T}$. Then, the one-step (conditional) distortion risk measure with distortion functions $\{g_t\}_{t\in\mathcal{T}}$ is the family $\{\rho_t\}_{t\in\mathcal{T}}$, where for each $t \in \mathcal{T}$ and $Z \in \mathcal{Z}_{t+1}$, ρ_t is defined as

$$\rho_t(Z) := -\int_{-\infty}^{0} [1 - g_t(1 - F_{Z|F_t}(x))]\, dx + \int_0^{+\infty} g_t(1 - F_{Z|F_t}(x))\, dx.$$

If $g_t(\cdot,\omega)$ is absolutely continuous for all $\omega \in \Omega$, then the one-step risk measure admits representation (2), where the distortion weight function $\gamma_t : (0,1) \times \Omega \to \mathbb{R}_+$ is given by $\gamma_t(u,\omega) := \frac{\partial^-}{\partial x} g(x,\omega)|_{x=1-u}$, with $\frac{\partial^-}{\partial x}$ being the left derivative with respect to x for all $u \in (0,1)$ and $\omega \in \Omega$; see, for example, Dhaene et al. (2012) for a proof in the static setting.

Lemma A.1. *Under Assumption 1, any coherent one-step distortion risk measure ρ_t is a mapping $\rho_t : \mathcal{Z}_{t+1} \to \mathcal{Z}_t$, $t \in \mathcal{T}/\{0\}$ and $\rho_0 : \mathcal{Z}_1 \to \mathbb{R}$.*

Proof. Take $Z \in \mathcal{Z}_{t+1}$; then, we have by (2) that $\rho_t(Z)$ is $\mathcal{F}_t$ measurable and that

$$\begin{aligned}\mathbb{E}[|\rho_t(Z)|^2] &= \mathbb{E}[|\mathbb{E}[Z\,\gamma_t(U_{Z|\mathcal{F}_t})|\mathcal{F}_t]|^2] \\ &\le \mathbb{E}[(\mathbb{E}[Z^2|\mathcal{F}_t])(\mathbb{E}[(\gamma_t(U_{Z|\mathcal{F}_t})^2|\mathcal{F}_t])] \\ &\le C\,\mathbb{E}[\mathbb{E}[Z^2|\mathcal{F}_t]] = C\,\mathbb{E}[Z^2] < +\infty,\end{aligned}$$

where the first inequality follows from Cauchy–Schwartz, the second inequality follows from Assumption 1, and the third inequality follows from $Z \in \mathcal{Z}_{t+1}$. □

Lemma A.2. *For any $c \ge 0$, the set of admissible strategies $\mathcal{A}_c$ is a convex set.*

Proof. Let $\boldsymbol{\theta}^{(0)}$ and $\boldsymbol{\theta}^{(1)}$ be two admissible strategies, and define $\boldsymbol{\theta}^{(a)} := a\,\boldsymbol{\theta}^{(0)} + (1-a)\,\boldsymbol{\theta}^{(1)}$ for $a \in [0,1]$. First, note that $\boldsymbol{\theta}^{(a)} \in \boldsymbol{\mathcal{Z}}_{0:T}$.

Part 1. $\boldsymbol{\theta}^{(a)}_{0:T}$ is lower bounded by c. Indeed, for $t \in \mathcal{T}$, when $c > 0$, $\boldsymbol{\theta}^{(a)}_t = a\,\boldsymbol{\theta}^{(0)}_t + (1-a)\,\boldsymbol{\theta}^{(1)}_t \ge ac + (1-a)c = c$, and when $c = 0$, $\boldsymbol{\theta}^{(a)}_t = a\,\boldsymbol{\theta}^{(0)}_t + (1-a)\,\boldsymbol{\theta}^{(1)}_t > 0$.

Part 2. It holds that $w^{\boldsymbol{\theta}^{(a)}} \in \mathcal{L}^{\infty}_{t+1}$. For this, note that

$$\begin{aligned}w^{\boldsymbol{\theta}^{(a)}}_t = \frac{\boldsymbol{\theta}^{(a)\top}_t \boldsymbol{X}_{t+1}}{\boldsymbol{\theta}^{(a)\top}_{t+1} \boldsymbol{X}_{t+1}} &= a\,\frac{\boldsymbol{\theta}^{(0)\top}_t \boldsymbol{X}_{t+1}}{(a\,\boldsymbol{\theta}^{(0)}_{t+1} + (1-a)\,\boldsymbol{\theta}^{(1)}_{t+1})^\top \boldsymbol{X}_{t+1}} \\ &\quad + (1-a)\frac{\boldsymbol{\theta}^{(1)\top}_t \boldsymbol{X}_{t+1}}{(a\,\boldsymbol{\theta}^{(0)}_{t+1} + (1-a)\,\boldsymbol{\theta}^{(1)}_{t+1})^\top \boldsymbol{X}_{t+1}} \\ &\le a\,\frac{\boldsymbol{\theta}^{(0)\top}_t \boldsymbol{X}_{t+1}}{a\,\boldsymbol{\theta}^{(0)\top}_{t+1} \boldsymbol{X}_{t+1}} + (1-a)\,\frac{\boldsymbol{\theta}^{(1)\top}_t \boldsymbol{X}_{t+1}}{(1-a)\,\boldsymbol{\theta}^{(1)\top}_{t+1} \boldsymbol{X}_{t+1}} \\ &= w^{\boldsymbol{\theta}^{(0)}}_t + w^{\boldsymbol{\theta}^{(1)}}_t < +\infty,\end{aligned}$$

where the first inequality follows as $\boldsymbol{\theta}^{(i)\top}_t \boldsymbol{X}_t \ge 0$ and the last inequality follows by admissibility of $\boldsymbol{\theta}^{(0)}$ and $\boldsymbol{\theta}^{(1)}$.

Part 3. For every $t \in \mathcal{T}$, $i \in \mathcal{N}$, it holds that

$$\begin{aligned}(\mathbb{E}[(\theta^{(a)}_{t,i}\Delta X_{t,i})^2])^{\frac{1}{2}} &= (\mathbb{E}[((a\,\theta^{(0)}_{t,i} + (1-a)\,\theta^{(1)}_{t,i})\Delta X_{t,i})^2])^{\frac{1}{2}} \\ &\le (\mathbb{E}[(a\,\theta^{(0)}_{t,i}\Delta X_{t,i})^2])^{\frac{1}{2}} + (\mathbb{E}[((1-a)\,\theta^{(1)}_{t,i}\Delta X_{t,i})^2])^{\frac{1}{2}} \\ &< +\infty,\end{aligned}$$

where the inequality is the triangle inequality and the strict inequality by admissibility of $\boldsymbol{\theta}^{(0)}$ and $\boldsymbol{\theta}^{(1)}$. Hence, we have that $\sum_{t\in\mathcal{T}, i\in\mathcal{N}} \mathbb{E}[(\theta^{(a)}_{t,i}\Delta X_{t,i})^2] < +\infty$.

Parts (1)–(3) imply that $\boldsymbol{\theta}^{(a)}$ is admissible. □

Lemma A.3. *For any $c \ge 0$, if the risk measure is a coherent distortion DRM satisfying Assumption 1, then any admissible strategy $\boldsymbol{\theta} \in \mathcal{A}_c$ satisfies $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] \in \mathcal{Z}_t$, $t \in \mathcal{T}/\{0\}$, and $\mathfrak{R}[\boldsymbol{\theta}_{0:T}] < +\infty$.*

Proof. Recall that for all $t \in \mathcal{T}/\{0\}$, the one-step distortion risk measures are mappings $\rho_t : \mathcal{Z}_{t+1} \to \mathcal{Z}_t$. Next, we proceed by induction backward in time. At time T, the risk is

$$\mathfrak{R}_T[\boldsymbol{\theta}_T] = \rho_T(\boldsymbol{\theta}^\top_T \Delta \boldsymbol{X}_T).$$

By admissibility of $\boldsymbol{\theta}$, we have $\boldsymbol{\theta}^\top_T \Delta \boldsymbol{X}_T \in \mathcal{Z}_{T+1}$; thus, $\mathfrak{R}_T[\boldsymbol{\theta}_T] \in \mathcal{Z}_T$. Next, assume that $\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1}] \in \mathcal{Z}_{t+1}$; then, the risk-to-go at time t is

$$\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] = \rho_t(\boldsymbol{\theta}^\top_t \Delta \boldsymbol{X}_t + w^{\boldsymbol{\theta}}_t \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]).$$

By admissibility of $\boldsymbol{\theta}$, we have $\boldsymbol{\theta}^\top_t \Delta \boldsymbol{X}_t \in \mathcal{Z}_{t+1}$, $w^{\boldsymbol{\theta}}_t \in \mathcal{L}^{\infty}_{t+1}$, and thus, $w^{\boldsymbol{\theta}}_t \mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}] \in \mathcal{Z}_{t+1}$, which implies that $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}] \in \mathcal{Z}_t$. A similar argument yields that $\mathfrak{R}[\boldsymbol{\theta}_{0:T}] < +\infty$. □

Lemma A.4. *For any $c \ge 0$ and $\boldsymbol{\theta}_{0:T} \in \mathcal{A}_c$, we have that $RC_{t,i}[\boldsymbol{\theta}_{t:T}] \in \mathcal{Z}_t$ for all $t \in \mathcal{T}$ and $i \in \mathcal{N}$.*

Proof. Note that

$$
\begin{aligned}
&|RC_{t,i}[\boldsymbol{\theta}_{t:T}]|^2 \\
&\le \left|\mathbb{E}\left[\theta_{t,i}\Delta X_{t,i}\gamma_t(U_t[\boldsymbol{\theta}_{t:T}]) + \frac{\theta_{t,i}X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top \boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]\gamma_t(U_t[\boldsymbol{\theta}_{t:T}])\,\middle|\,\mathcal{F}_t\right]\right|^2 \\
&\le 2|\mathbb{E}[\theta_{t,i}\Delta X_{t,i}\gamma_t(U_t[\boldsymbol{\theta}_{t:T}])|\mathcal{F}_t]|^2 \\
&\quad + 2\left|\mathbb{E}\left[\frac{\theta_{t,i}X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top \boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]\gamma_t(U_t[\boldsymbol{\theta}_{t:T}])\,\middle|\,\mathcal{F}_t\right]\right|^2 \\
&\le 2\mathbb{E}[(\theta_{t,i}\Delta X_{t,i})^2|\mathcal{F}_t]\mathbb{E}[(\gamma_t(U_t[\boldsymbol{\theta}_{t:T}]))^2|\mathcal{F}_t] \\
&\quad + 2\mathbb{E}\left[\left(\frac{\theta_{t,i}X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top \boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]\right)^2\,\middle|\,\mathcal{F}_t\right]\mathbb{E}[(\gamma_t(U_t[\boldsymbol{\theta}_{t:T}]))^2|\mathcal{F}_t] \\
&\le 2C\mathbb{E}[(\theta_{t,i}\Delta X_{t,i})^2|\mathcal{F}_t] + 2C\mathbb{E}\left[\left(\frac{\theta_{t,i}X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top \boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}]\right)^2\,\middle|\,\mathcal{F}_t\right] \\
&\le 2C\mathbb{E}\left[\sum_{t\in\mathcal{T},i\in\mathcal{N}}(\theta_{t,i}\Delta X_{t,i})^2|\mathcal{F}_t\right] + 2C\mathbb{E}[(w_t^{\boldsymbol{\theta}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}])^2\,|\,\mathcal{F}_t],
\end{aligned}
$$

and hence,

$$
\begin{aligned}
\mathbb{E}[|RC_{t,i}[\boldsymbol{\theta}_{t:T}]|^2] &\le 2C\mathbb{E}\left[\sum_{t\in\mathcal{T},i\in\mathcal{N}}(\theta_{t,i}\Delta X_{t,i})^2\right] \\
&\quad + 2C\mathbb{E}[(w_t^{\boldsymbol{\theta}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}])^2] < +\infty,
\end{aligned}
$$

where the last inequality follows as (i) $w_t^{\boldsymbol{\theta}}$ are in $\mathcal{L}^\infty$, (ii) $\boldsymbol{\theta}$ is admissible, and (iii) $\mathfrak{R}_{t+1}[\boldsymbol{\theta}_{t+1:T}] \in \mathcal{Z}_{t+1}$ from Lemma A.2. □

Proposition A.1 (Lipschitz Continuity of One-Step Coherent Distortion Risk Measures). *Let ρ_t be a one-step coherent distortion risk measure with distortion weight function γ_t satisfying Assumption 1. Then, ρ_t is Lipschitz continuous with respect to the conditional L^2 norm (i.e., for any $X, Y \in \mathcal{Z}_{t+1}$, it holds that*

$$
|\rho_t(X) - \rho_t(Y)| \le C^{\frac{1}{2}}\|X - Y\|_t \quad a.s.,
$$

where the conditional norm $\|\cdot\|_t$ is defined for all $Z \in \mathcal{Z}_{t+1}$ by $\|Z\|_t := (\mathbb{E}[Z^2|\mathcal{F}_t])^{\frac{1}{2}}$, $\forall t \in \mathcal{T}/\{0\}$). Furthermore, $\rho_0 : \mathcal{Z}_1 \to \mathbb{R}$.

Proof. This is an adaption of Inoue (2003, lemma 2.1) to one-step distortion risk measures. By subadditivity of the conditional distortion risk measure, we have a.s.

$$
\begin{aligned}
\rho_t(X) - \rho_t(Y) \le \rho_t(X - Y) &= \mathbb{E}[(X - Y)\gamma_t(U_{X-Y|\mathcal{F}_t})|\mathcal{F}_t] \\
&\le (\mathbb{E}[(X - Y)^2|\mathcal{F}_t])^{\frac{1}{2}}(\mathbb{E}[(\gamma_t(U_{X-Y|\mathcal{F}_t}))^2|\mathcal{F}_t])^{\frac{1}{2}} \\
&\le C^{\frac{1}{2}}\|X - Y\|_t.
\end{aligned}
$$

In the second inequality, we use the conditional Cauchy–Schwarz inequality, and the last inequality follows by assumption on γ_t. Interchanging X and Y concludes the proof.

The fact that $\rho_0 : Z_1 \to \mathbb{R}$ follows by, for example, Shapiro et al. (2021, definition 6.37). □

Proposition A.2 (Interchanging Expectation and Limits). *Suppose that $\boldsymbol{\theta} = (\boldsymbol{\theta}_t, \boldsymbol{\theta}^*_{t+1:T})$ and $\boldsymbol{\theta}' = (\boldsymbol{\theta}'_t, \boldsymbol{\theta}^*_{t+1:T})$ are admissible strategies, and let $\delta\boldsymbol{\theta} := \boldsymbol{\theta}'_t - \boldsymbol{\theta}_t$; then, we have that*

$$
\begin{aligned}
&\lim_{\varepsilon\downarrow 0}\frac{1}{\varepsilon}(L_t[\boldsymbol{\theta}_t + \varepsilon\delta\boldsymbol{\theta}] - L_t[\boldsymbol{\theta}_t]) \\
&= \sum_{i\in\mathcal{N}}\mathbb{E}\left[\mathcal{D}_i^{\delta\theta_i}\mathfrak{R}_t[(\boldsymbol{\theta}_t, \boldsymbol{\theta}^*_{t+1:T})] - b_{t,i}\frac{\delta\theta_i}{\theta_{t,i}}\right].
\end{aligned}
$$

Proof. For any $\varepsilon \in [0,1]$, let $\boldsymbol{\theta}^\varepsilon := \boldsymbol{\theta} + \varepsilon\,\delta\boldsymbol{\theta} = (1-\varepsilon)\,\boldsymbol{\theta} + \varepsilon\,\boldsymbol{\theta}'$. By Lemma A.2, $\boldsymbol{\theta}^\varepsilon$ is admissible. Then, by definition, we have that

$$
\begin{aligned}
&\lim_{\varepsilon\downarrow 0}\frac{1}{\varepsilon}(L_t[\boldsymbol{\theta}_t + \varepsilon\delta\boldsymbol{\theta}] - L_t[\boldsymbol{\theta}_t]) \\
&= \lim_{\varepsilon\downarrow 0}\frac{1}{\varepsilon}\mathbb{E}\left[\mathfrak{R}_t[(\boldsymbol{\theta}_t^\varepsilon, \boldsymbol{\theta}^*_{t+1:T})] - \mathfrak{R}_t[(\boldsymbol{\theta}_t, \boldsymbol{\theta}^*_{t+1:T})] - \sum_{i\in\mathcal{N}} b_{t,i}(\log\theta^\varepsilon_{t,i} - \log\theta_{t,i})\right] \\
&= \lim_{\varepsilon\downarrow 0}\frac{1}{\varepsilon}\mathbb{E}[\underbrace{\rho_t(\boldsymbol{\theta}_t^{\varepsilon\top}\Delta\boldsymbol{X}_t + w_t^{\boldsymbol{\theta}^\varepsilon}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]) - \rho_t(\boldsymbol{\theta}_t^\top\Delta\boldsymbol{X}_t + w_t^{\boldsymbol{\theta}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}])}_{=:A_t} \\
&\qquad \underbrace{- \sum_{i\in\mathcal{N}\setminus b_{t,i}}(\log\theta^\varepsilon_{t,i} - \log\theta_{t,i})}_{=:B_t}].
\end{aligned} \tag{A.1}
$$

We next show that $\frac{1}{\varepsilon}|A_t| \le \mathfrak{A}_t$ s.t., $\mathfrak{A}_t$ is independent of ε and $\mathbb{E}[\mathfrak{A}_t] < +\infty$, and similarly, for $\frac{1}{\varepsilon}|B_t| \le \mathfrak{B}_t$ s.t., $\mathfrak{B}_t$ is independent of ε and $\mathbb{E}[\mathfrak{B}_t] < +\infty$.

To this end, by Proposition A.1, we have that

$$
\begin{aligned}
\frac{1}{\varepsilon}|A_t| &\le \frac{1}{\varepsilon}C^{\frac{1}{2}}\|\varepsilon\delta\boldsymbol{\theta}^\top\Delta\boldsymbol{X}_t + (w_t^{\boldsymbol{\theta}^\varepsilon} - w_t^{\boldsymbol{\theta}})\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\|_t \\
&= C^{\frac{1}{2}}\left\|\delta\boldsymbol{\theta}^\top\Delta\boldsymbol{X}_t + \frac{\delta\boldsymbol{\theta}^\top\boldsymbol{X}_{t+1}}{\boldsymbol{\theta}^{*\top}_{t+1}\boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\right\|_t \\
\text{(by } \triangle \text{ inequality)} &\le C^{\frac{1}{2}}\left(\|\delta\boldsymbol{\theta}^\top\Delta\boldsymbol{X}_t\|_t + \left\|\frac{\delta\boldsymbol{\theta}^\top\boldsymbol{X}_{t+1}}{\boldsymbol{\theta}^{*\top}_{t+1}\boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\right\|_t\right) =: \mathfrak{A}_t \qquad \text{a.s.}
\end{aligned}
$$

Note that $\mathfrak{A}_t$ is independent of ε; furthermore,

$$
\begin{aligned}
&C^{-\frac{1}{2}}\mathbb{E}[\mathfrak{A}_t] \\
&\le (\mathbb{E}[\mathbb{E}[(\delta\boldsymbol{\theta}^\top\Delta\boldsymbol{X}_t)^2|\mathcal{F}_t]])^{\frac{1}{2}} \quad \text{(by definition } \|\cdot\|_t \text{ and Jensen)} \\
&\quad + \left(\mathbb{E}\left[\mathbb{E}\left[\left(\frac{\delta\boldsymbol{\theta}^\top\boldsymbol{X}_{t+1}}{\boldsymbol{\theta}^{*\top}_{t+1}\boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\right)^2|\mathcal{F}_t\right]\right]\right)^{\frac{1}{2}} \\
&= \|\delta\boldsymbol{\theta}^\top\Delta\boldsymbol{X}_t\|_0 + \left\|\frac{\delta\boldsymbol{\theta}^\top\boldsymbol{X}_{t+1}}{\boldsymbol{\theta}^{*\top}_{t+1}\boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\right\|_0 \\
&\text{(by } \triangle \text{ inequality)} \\
&\le \|\boldsymbol{\theta}_t^\top\Delta\boldsymbol{X}_t\|_0 + \|\boldsymbol{\theta}_t'^\top\Delta\boldsymbol{X}_t\|_0 + \left\|\frac{\boldsymbol{\theta}_t^\top\boldsymbol{X}_{t+1}}{\boldsymbol{\theta}^{*\top}_{t+1}\boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\right\|_0 \\
&\quad + \left\|\frac{\boldsymbol{\theta}_t'^\top\boldsymbol{X}_{t+1}}{\boldsymbol{\theta}^{*\top}_{t+1}\boldsymbol{X}_{t+1}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\right\|_0 \\
&\le n^{\frac{1}{2}}\left(\sum_{i\in\mathcal{N}}\mathbb{E}[(\theta_{t,i}\Delta X_{t,i})^2]\right)^{\frac{1}{2}} + n^{\frac{1}{2}}\left(\sum_{i\in\mathcal{N}}\mathbb{E}[(\theta'_{t,i}\Delta X_{t,i})^2]\right)^{\frac{1}{2}} \\
&\quad + \|w^{\boldsymbol{\theta}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\|_0 + \|w^{\boldsymbol{\theta}'}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\|_0 \\
&\le n^{\frac{1}{2}}\left(\sum_{t\in\mathcal{T},i\in\mathcal{N}}\mathbb{E}[(\theta_{t,i}\Delta X_{t,i})^2]\right)^{\frac{1}{2}} + n^{\frac{1}{2}}\left(\sum_{t\in\mathcal{T},i\in\mathcal{N}}\mathbb{E}[(\theta'_{t,i}\Delta X_{t,i})^2]\right)^{\frac{1}{2}} \\
&\quad + \|w^{\boldsymbol{\theta}}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\|_0 + \|w^{\boldsymbol{\theta}'}\mathfrak{R}_{t+1}[\boldsymbol{\theta}^*_{t+1:T}]\|_0 \\
&< +\infty,
\end{aligned}
$$

where the last inequality follows from the admissibility of $\boldsymbol{\theta}$ and $\boldsymbol{\theta}'$ and from Lemma A.3.

Next, as $|\log(x)-\log(y)| \le \frac{1}{c}|x-y|$ for all $x,y \ge c > 0$, we have that

$$\left|\frac{1}{\varepsilon}B_t\right| \le \sum_{i\in\mathcal{N}} \frac{c}{\varepsilon}|\theta^{\varepsilon}_{t,i}-\theta_{t,i}| = c\sum_{i\in\mathcal{N}}|\delta\theta_{t,i}| \le \sum_{i\in\mathcal{N}} c(|\theta'_{t,i}|+|\theta_{t,i}|) =: \mathfrak{B}_t,$$

where c is the lower bound for admissible strategies and $\mathfrak{B}_t$ is independent of ε. As $\boldsymbol{\theta}$ and $\boldsymbol{\theta}'$ are in $\boldsymbol{\mathcal{Z}}$, we have that $\mathbb{E}[\mathfrak{B}_t] < +\infty$.

Putting these bounds together, we have by Lebesgue dominated convergence that the $\lim_{\varepsilon\downarrow 0}$ in (A.1) may be moved under the expectation, and the claim follows. □

Proposition A.3. *Let ρ_t be a one-step coherent distortion risk measure with distortion weight function γ_t satisfying Assumption 1. Let $t \in \mathcal{T}$ and $Y, W \in \mathcal{Z}_{t+1}$, where we assume that (Y, W) has a joint density, although the proof can be generalized to include point masses. If $\varepsilon \to F^{-1}_{Y+\varepsilon W}(u)$ is differentiable in a neighborhood around $\varepsilon = 0$ with bounded derivative for all $u \in (0,1)$, then it holds that*

$$\lim_{\varepsilon\to 0}\frac{\rho_t(Y+\varepsilon W)-\rho_t(Y)}{\varepsilon} = \mathbb{E}[W\gamma_t(U_{Y|\mathcal{F}_t})|\mathcal{F}_t]. \tag{A.2}$$

Proof. First, we define the conditional cdfs $F(y) := \mathbb{P}(Y \le y|\mathcal{F}_t)$ and $F(y,\varepsilon) := \mathbb{P}(Y+\varepsilon W \le y|\mathcal{F}_t)$ and their corresponding densities by $f(y)$ and $f(y,\varepsilon)$. We further write $F^{-1}(u)$ and $F^{-1}(u,\varepsilon)$ for the quantile functions of $F(\cdot)$ and $F(\cdot,\varepsilon)$, respectively.

Next, using $\rho_t(Y+\varepsilon W) = \mathbb{E}[F^{-1}(U,\varepsilon)\gamma_t(U)|\mathcal{F}_t]$ for a uniform rv $U \in \mathcal{F}_t$, the integrability assumption on γ_t, and the differentiability assumption on $F^{-1}(u,\varepsilon)$, the mean value theorem together with Lebesgue dominated convergence allows us to interchange expectation and limit to obtain

$$\lim_{\varepsilon\to 0}\frac{\rho_t(Y+\varepsilon W)-\rho_t(Y)}{\varepsilon} = \mathbb{E}[\partial_\varepsilon F^{-1}(U,\varepsilon)\gamma_t(U)|\mathcal{F}_t]|_{\varepsilon=0}. \tag{A.3}$$

By taking a derivative with respect to ε of the equation $F(F^{-1}(u,\varepsilon),\varepsilon) = u$, we obtain for all $u \in (0,1)$ that

$$\partial_\varepsilon F^{-1}(u,\varepsilon) = -\frac{\partial_\varepsilon F(y,\varepsilon)}{f(y,\varepsilon)}\bigg|_{y=F^{-1}(u,\varepsilon)}. \tag{A.4}$$

Next, we calculate the derivative $\partial_\varepsilon F(y,\varepsilon)$. For this, note that

$$\begin{aligned}
\partial_\varepsilon F(y,\varepsilon) &= \partial_\varepsilon \mathbb{E}[\mathbb{1}_{Y+\varepsilon W\le y}|\mathcal{F}_t] = \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\mathbb{E}[\mathbb{1}_{Y+\varepsilon W\le y} - \mathbb{1}_{Y\le y}|\mathcal{F}_t] \\
&= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\mathbb{E}[\mathbb{E}[\mathbb{1}_{Y\in(y-\varepsilon W,y]}|W]|\mathcal{F}_t] \\
&= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\mathbb{E}\left[\int_{y-\varepsilon W}^{y} dF_{Y|W}(y')|\mathcal{F}_t\right] \\
&= -\mathbb{E}[W f_{Y|W}(y)|\mathcal{F}_t],
\end{aligned} \tag{A.5}$$

where $F_{Y|W}$ and $f_{Y|W}$ are the distribution and density, respectively, of Y conditional on W. Plugging (A.4) and (A.5) into Equation (A.3), we obtain

$$\begin{aligned}
\lim_{\varepsilon\to 0}\frac{\rho_t(Y+\varepsilon W)-\rho_t(Y)}{\varepsilon} &= \mathbb{E}\left[\frac{\mathbb{E}[W f_{Y|W}(y)]}{f(y)}\bigg|_{y=F^{-1}(U)}\gamma_t(U)\,\bigg|\,\mathcal{F}_t\right] \\
&= \mathbb{E}[\mathbb{E}[W|Y=y]|_{y=F^{-1}(U)}\gamma_t(U)\,|\,\mathcal{F}_t] \\
&= \mathbb{E}[\mathbb{E}[W|F(Y)=U]\gamma_t(U)\,|\,\mathcal{F}_t] \\
&= \mathbb{E}[W\gamma_t(U_{Y|\mathcal{F}_t})|\mathcal{F}_t]. \quad \square
\end{aligned}$$

Appendix B. Additional Proofs

B.1. Proof of Proposition 4

Proof of Proposition 4. First, we generalize the Gâteaux derivative as follows. For a functional $F_t : \boldsymbol{\mathcal{Z}}_{t:T} \to \mathcal{Z}_t$, $t \in \mathcal{T}$, and $s \ge t$, we denote by $\mathcal{D}^{\zeta}_{s,i}F_t$ its Gâteaux derivative of the ith component at time s in direction $\zeta \in \mathcal{Z}_s$. That is, for $s,t \in \mathcal{T}$, $s \ge t$, and $\boldsymbol{Z}_{t:T} \in \boldsymbol{\mathcal{Z}}$,

$$\mathcal{D}^{\zeta}_{s,i}F_t[\boldsymbol{Z}_{t:T}] := \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}(F_t[\boldsymbol{Z}_{t:T}+\varepsilon\mathbf{1}_{s,i}\zeta] - F_t[\boldsymbol{Z}_{t:T}]).$$

Next, note that

$$\begin{aligned}
RC_{t,i}[\boldsymbol{\theta}_{t:T}] &= \mathcal{D}^{\theta_{t,i}}_i \rho_t(\boldsymbol{\theta}_t^\top \Delta\boldsymbol{X}_t + \mathfrak{R}_{t+1}[w_t^{\boldsymbol{\theta}}\boldsymbol{\theta}_{t+1:T}]) \\
&= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\{\rho_t(\varepsilon\theta_{t,i}\Delta X_{t,i} + \varepsilon\mathcal{D}^{\theta_{t,i}}_{t,i}\mathfrak{R}_{t+1}[w_t^{\boldsymbol{\theta}}\boldsymbol{\theta}_{t+1:T}] \\
&\quad + \boldsymbol{\theta}_t^\top\Delta\boldsymbol{X}_t + \mathfrak{R}_{t+1}[w_t^{\boldsymbol{\theta}}\boldsymbol{\theta}_{t+1:T}]) - \mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]\} \\
&\overset{\text{by (9)}}{=} \mathbb{E}[(\theta_{t,i}\Delta X_{t,i} + \mathcal{D}^{\theta_{t,i}}_{t,i}\mathfrak{R}_{t+1}[w_t^{\boldsymbol{\theta}}\boldsymbol{\theta}_{t+1:T}])\Gamma_t^{\boldsymbol{\theta}}|\mathcal{F}_t].
\end{aligned} \tag{B.1}$$

Next, we show that for all $s \in \{t,\dots,T\}$,

$$\mathcal{D}^{\theta_{t,i}}_{t,i}\mathfrak{R}_s[w_t^{\boldsymbol{\theta}}\boldsymbol{\theta}_{s:T}] = \sum_{r=s}^{T}\mathbb{E}\left[\frac{\theta_{t,i}X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top\boldsymbol{X}_{t+1}}(\boldsymbol{\theta}_r^\top\Delta\boldsymbol{X}_r)\Gamma_s^{\boldsymbol{\theta}}\cdots\Gamma_r^{\boldsymbol{\theta}}\Big|\mathcal{F}_s\right]. \tag{B.2}$$

Although we require the above equation for the case $s = t+1$ only, to make the proof easier to understand, we introduce an additional variable s and use mathematical induction over s. First, we show that Equation (B.2) holds for $s = T$. To see this, we calculate

$$\begin{aligned}
\mathcal{D}^{\theta_{t,i}}_{t,i}\mathfrak{R}_T[w_t^{\boldsymbol{\theta}}\boldsymbol{\theta}_T] &= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\left\{\rho_T\left(w_t^{\boldsymbol{\theta}}(\boldsymbol{\theta}_T^\top\Delta\boldsymbol{X}_T) + \varepsilon\frac{\theta_{t,i}X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top\boldsymbol{X}_{t+1}}(\boldsymbol{\theta}_T^\top\Delta\boldsymbol{X}_T)\right)\right. \\
&\quad \left. - \rho_T\left(w_t^{\boldsymbol{\theta}}(\boldsymbol{\theta}_T^\top\Delta\boldsymbol{X}_T)\right)\right\} \\
&= \mathbb{E}\left[\frac{\theta_{t,i}X_{t+1,i}}{\boldsymbol{\theta}_{t+1}^\top\boldsymbol{X}_{t+1}}(\boldsymbol{\theta}_T^\top\Delta\boldsymbol{X}_T)\Gamma_T^{\boldsymbol{\theta}}|\mathcal{F}_T\right],
\end{aligned}$$

where we applied Equation (9) to obtain the last equality. Next, assuming that Equation (B.2) holds for all $r \in \{s+1,\dots,T\}$, we show that it also holds for s. Indeed, using the definition of a Gâteaux derivative (e.g., that $F_s[\boldsymbol{Z}_{s:T}+$

$\varepsilon \mathbf{1}_{t,i}\zeta] = \varepsilon\, \mathcal{D}^{\zeta}_{t,i} F_s[\boldsymbol{Z}_{s:T}] + F_s[\boldsymbol{Z}_{s:T}] + o(\varepsilon))$, we obtain

$$
\begin{aligned}
&\mathcal{D}^{\theta_{t,i}}_{t,i}\, \Re_s[w^{\boldsymbol{\theta}}_t\, \boldsymbol{\theta}_{s:T}] \\
&\overset{\text{by (3)}}{=} \mathcal{D}^{\theta_{t,i}}_{t,i}\, \rho_s(w^{\boldsymbol{\theta}}_t\, \boldsymbol{\theta}^{\top}_s \Delta \boldsymbol{X}_s + \Re_{s+1}[w^{\boldsymbol{\theta}}_t\, \boldsymbol{\theta}_{s+1:T}]) \\
&= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \Bigg\{ \rho_s \Bigg(\varepsilon\, \frac{\theta^{\top}_{t,i} X_{t+1,i}}{\boldsymbol{\theta}^{\top}_{t+1} \boldsymbol{X}_{t+1}}\, \boldsymbol{\theta}^{\top}_s \Delta \boldsymbol{X}_s + w^{\boldsymbol{\theta}}_t\, \boldsymbol{\theta}^{\top}_s \Delta \boldsymbol{X}_s \\
&\qquad + \varepsilon\, \mathcal{D}^{\theta_{t,i}}_{t,i}\, \Re_{s+1}[w^{\boldsymbol{\theta}}_t\, \boldsymbol{\theta}_{s+1:T}] + \Re_{s+1}[w^{\boldsymbol{\theta}}_t\, \boldsymbol{\theta}_{s+1:T}] \Bigg) - \Re_s[w^{\boldsymbol{\theta}}_t\, \boldsymbol{\theta}_{s:T}] \Bigg\} \\
&\overset{\text{by (9)}}{=} \mathbb{E}\left[\left(\frac{\theta^{\top}_{t,i} X_{t+1,i}}{\boldsymbol{\theta}^{\top}_{t+1} \boldsymbol{X}_{t+1}}\, \boldsymbol{\theta}^{\top}_s \Delta \boldsymbol{X}_s + \mathcal{D}^{\theta_{t,i}}_{t,i}\, \Re_{s+1}[w^{\boldsymbol{\theta}}_t\, \boldsymbol{\theta}_{s+1:T}] \right) \Gamma^{\boldsymbol{\theta}}_s \middle| \mathcal{F}_s \right] \\
&= \mathbb{E}\left[\left(\frac{\theta^{\top}_{t,i} X_{t+1,i}}{\boldsymbol{\theta}^{\top}_{t+1} \boldsymbol{X}_{t+1}}\, \boldsymbol{\theta}^{\top}_s \Delta \boldsymbol{X}_s + \sum_{r=s+1}^{T} \mathbb{E}\left[\frac{\theta_{t,i} X_{t+1,i}}{\boldsymbol{\theta}^{\top}_{t+1} \boldsymbol{X}_{t+1}}\, \boldsymbol{\theta}^{\top}_r \right. \right. \right. \\
&\qquad \left. \left. \left. \Delta \boldsymbol{X}_r\, \Gamma^{\boldsymbol{\theta}}_{s+1} \cdots \Gamma^{\boldsymbol{\theta}}_r \middle| \mathcal{F}_{s+1} \right] \right) \Gamma^{\boldsymbol{\theta}}_s \middle| \mathcal{F}_s \right] \\
&= \sum_{r=s}^{T} \mathbb{E}\left[\mathbb{E}\left[\frac{\theta_{t,i} X_{t+1,i}}{\boldsymbol{\theta}^{\top}_{t+1} \boldsymbol{X}_{t+1}}\, \boldsymbol{\theta}^{\top}_r \Delta \boldsymbol{X}_r\, \Gamma^{\boldsymbol{\theta}}_{s+1} \cdots \Gamma^{\boldsymbol{\theta}}_r \middle| \mathcal{F}_{s+1} \right] \Gamma^{\boldsymbol{\theta}}_s \middle| \mathcal{F}_s \right] \\
&= \sum_{r=s}^{T} \mathbb{E}\left[\frac{\theta_{t,i} X_{t+1,i}}{\boldsymbol{\theta}^{\top}_{t+1} \boldsymbol{X}_{t+1}}\, \boldsymbol{\theta}^{\top}_r \Delta \boldsymbol{X}_r\, \Gamma^{\boldsymbol{\theta}}_s \cdots \Gamma^{\boldsymbol{\theta}}_r \middle| \mathcal{F}_s \right],
\end{aligned}
$$

where we use the induction argument in the fourth equation and that $\Gamma^{\boldsymbol{\theta}}_s \in \mathcal{F}_s$ in the last equality. This concludes the proof of Equation (B.2).

Finally, combining (B.2) with (B.1), noticing that $\Gamma^{\boldsymbol{\theta}}_t \in \mathcal{F}_t$, and using the law of iterated expectations conclude the proof. □

B.2. Proof of Theorem 4

Proof of Theorem 4. Let $\boldsymbol{\varphi}_{0:T} \in \mathcal{A}^{\mathrm{o}}_c$ denote a self-financing risk budgeting strategy with budget B and initial wealth of one whose risk-to-go satisfies $c_R \le \Re_t[\boldsymbol{\varphi}_{t:T}] \le c^R$ a.s. for all $t \in \mathcal{T}$. We show that $\boldsymbol{\varphi}_{0:T} = \frac{1}{\boldsymbol{\vartheta}^{*\top}_0 \boldsymbol{X}_0}\, \boldsymbol{\vartheta}^{*}_{0:T}$, where $\boldsymbol{\vartheta}^{*}_{0:T}$ is the induced self-financing strategy of the unique solution to optimization Problem (P) (i.e., given by Theorem 3(b) with lower bound $c' := \frac{c}{c^R}$). For this, we proceed by contradiction. Assume that $\boldsymbol{\varphi}_{0:T} \neq \frac{1}{\boldsymbol{\vartheta}^{*\top}_0 \boldsymbol{X}_0}\, \boldsymbol{\vartheta}^{*}_{0:T}$, and define the (not necessarily self-financing) strategy $\boldsymbol{\psi}_{0:T}$ via

$$
\boldsymbol{\psi}_t := \frac{1}{\Re_t[\boldsymbol{\varphi}_{t:T}]}\, \boldsymbol{\varphi}_t, \qquad \forall t \in \mathcal{T}. \tag{B.3}
$$

It holds that $\boldsymbol{\psi} \in \mathcal{A}^{\mathrm{o}}_{c'}$. This follows as the bounds on $\Re_t[\boldsymbol{\psi}_{t:T}]$ implies that $w^{\boldsymbol{\psi}}_t \in \mathcal{L}^{\infty}_t$, and $\Re_t[\boldsymbol{\psi}_{t:T}] \le c^R$ guarantees that $\boldsymbol{\psi}_t = \frac{1}{\Re_t[\boldsymbol{\varphi}_{t:T}]}\, \boldsymbol{\varphi}_t \ge c'$ for all $t \in \mathcal{T}$. Next, we show that the risk-to-go process of $\boldsymbol{\psi}_{0:T}$ satisfies

$$
\Re_t[\boldsymbol{\psi}_{t:T}] = 1, \qquad \forall t \in \mathcal{T}. \tag{B.4}
$$

We proceed by induction. At time T, by positive homogeneity of $\Re_T$ and because $c_R \le \Re_T[\boldsymbol{\varphi}_T] \le c^R$,

$$
\Re_T[\boldsymbol{\psi}_T] = \frac{1}{\Re_T[\boldsymbol{\varphi}_T]}\, \Re_T[\boldsymbol{\varphi}_T] = 1.
$$

Assume that Equation (B.4) holds for $t+1$, and then, again using $c_R \le \Re_t[\boldsymbol{\varphi}_{t:T}] \le c^R$,

$$
\begin{aligned}
\Re_t[\boldsymbol{\psi}_{t:T}] &= \rho_t(\boldsymbol{\psi}^{\top}_t \Delta \boldsymbol{X}_t + w^{\boldsymbol{\psi}}_t\, \Re_{t+1}[\boldsymbol{\psi}_{t+1:T}]) \\
&= \rho_t\left(\boldsymbol{\psi}^{\top}_t \Delta \boldsymbol{X}_t + \frac{\boldsymbol{\psi}^{\top}_t \boldsymbol{X}_{t+1}}{\boldsymbol{\psi}^{\top}_{t+1} \boldsymbol{X}_{t+1}} \right) \\
&= \rho_t\left(\frac{1}{\Re_t[\boldsymbol{\varphi}_{t:T}]}\, \boldsymbol{\varphi}^{\top}_t \Delta \boldsymbol{X}_t + \frac{\Re_{t+1}[\boldsymbol{\varphi}_{t+1:T}]}{\Re_t[\boldsymbol{\varphi}_{t:T}]}\, \frac{\boldsymbol{\varphi}^{\top}_t \boldsymbol{X}_{t+1}}{\boldsymbol{\varphi}^{\top}_{t+1} \boldsymbol{X}_{t+1}} \right)
\end{aligned}
$$

(by positive homogeneity of $\rho_t(\cdot)$ and φ is self-financing)

$$
\begin{aligned}
&= \frac{1}{\Re_t[\boldsymbol{\varphi}_{t:T}]}\, \rho_t(\boldsymbol{\varphi}^{\top}_t \Delta \boldsymbol{X}_t + \Re_{t+1}[\boldsymbol{\varphi}_{t+1:T}]) \\
&= 1.
\end{aligned}
$$

Thus, Equation (B.4) holds for all $t \in \mathcal{T}$.

Next, we show that $\boldsymbol{\psi}_{0:T}$ is a risk budgeting strategy with budget B. By positive homogeneity of risk contributions (Proposition 3), we obtain

$$
\begin{aligned}
RC_{t,i}[\boldsymbol{\psi}_{t:T}] &= \frac{1}{\Re_t[\boldsymbol{\varphi}_{t:T}]}\, RC_{t,i}[\boldsymbol{\varphi}_{t:T}] = \frac{1}{\Re_t[\boldsymbol{\varphi}_{t:T}]}\, b_{t,i}\, \Re_t[\boldsymbol{\varphi}_{t:T}] = b_{t,i} \\
&= b_{t,i}\, \Re_t[\boldsymbol{\psi}_{t:T}].
\end{aligned}
$$

Thus, $\boldsymbol{\psi}_{0:T}$ is not only a risk budgeting strategy but also, a solution to optimization Problem (P) with lower bound c'; that is, for all $t \in \mathcal{T}$, it satisfies Equation (15). As $\boldsymbol{\psi} \in \mathcal{A}^{\mathrm{o}}_{c'}$ is a solution to optimization Problem (P) with $\mathcal{A}_{c'}$, it induces a self-financing risk budgeting strategy with initial wealth of one as given in Theorem 3(b) and denoted here by $\boldsymbol{\vartheta}_{0:T}$. Specifically, we have

$$
\boldsymbol{\vartheta}_0 := \frac{1}{\boldsymbol{\psi}^{\top}_0 \boldsymbol{X}_0}\, \boldsymbol{\psi}_0 \quad \text{and} \quad \boldsymbol{\vartheta}_t := \frac{1}{\boldsymbol{\psi}^{\top}_0 \boldsymbol{X}_0} \left(\prod_{s=0}^{t-1} w^{\boldsymbol{\psi}}_s \right) \boldsymbol{\psi}_t, \quad \forall t \in \mathcal{T}/\{0\}.
$$

Finally, we show that $\boldsymbol{\vartheta}_{0:T} = \boldsymbol{\varphi}_{0:T}$. For this, recall that $\boldsymbol{\varphi}_{0:T}$ is a self-financing strategy with initial wealth of one, and thus,

$$
\boldsymbol{\vartheta}_0 = \frac{1}{\boldsymbol{\psi}^{\top}_0 \boldsymbol{X}_0}\, \boldsymbol{\psi}_0 = \frac{\Re_0[\boldsymbol{\varphi}_{0:T}]}{\boldsymbol{\varphi}^{\top}_0 \boldsymbol{X}_0}\, \frac{\boldsymbol{\varphi}_0}{\Re_0[\boldsymbol{\varphi}_{0:T}]} = \boldsymbol{\varphi}_0.
$$

For $t \in \mathcal{T}/\{0\}$, we have

$$
\begin{aligned}
\boldsymbol{\vartheta}_t &= \frac{1}{\boldsymbol{\psi}^{\top}_0 \boldsymbol{X}_0} \left(\prod_{s=0}^{t-1} \frac{\boldsymbol{\psi}^{\top}_s \boldsymbol{X}_{s+1}}{\boldsymbol{\psi}^{\top}_{s+1} \boldsymbol{X}_{t+1}} \right) \boldsymbol{\psi}_t \\
&= \frac{\Re_0[\boldsymbol{\varphi}_{0:T}]}{\boldsymbol{\varphi}^{\top}_0 \boldsymbol{X}_0} \left(\prod_{s=0}^{t-1} \frac{\Re_{s+1}[\boldsymbol{\varphi}_{s+1:T}]}{\Re_s[\boldsymbol{\varphi}_{s:T}]}\, \frac{\boldsymbol{\varphi}^{\top}_s \boldsymbol{X}_{s+1}}{\boldsymbol{\varphi}^{\top}_{s+1} \boldsymbol{X}_{t+1}} \right) \frac{\boldsymbol{\varphi}_t}{\Re_t[\boldsymbol{\varphi}_{t:T}]}
\end{aligned}
$$

(as $\boldsymbol{\varphi}_{0:T}$ is self-financing)

$$
\begin{aligned}
&= \Re_0[\boldsymbol{\varphi}_{0:T}] \left(\prod_{s=0}^{t-1} \frac{\Re_{s+1}[\boldsymbol{\varphi}_{s+1:T}]}{\Re_s[\boldsymbol{\varphi}_{s:T}]} \right) \frac{\boldsymbol{\varphi}_t}{\Re_t[\boldsymbol{\varphi}_{t:T}]} \\
&= \boldsymbol{\varphi}_t.
\end{aligned}
$$

Thus, $\boldsymbol{\varphi}_{0:T} = \boldsymbol{\vartheta}_{0:T}$, and as $\boldsymbol{\vartheta}_{0:T}$ is given by Theorem 3(b), we arrive at a contradiction. Uniqueness follows from strict convexity of the optimization Problem (P). □

Appendix C. Elicitability

We first recall the classical definition of elicitability and scoring functions; see, for example, Fissler and Ziegel (2016). For this, first define the set of cdfs $\mathcal{M} := \{F_Y | Y \in \mathcal{F}_{T+1}\}$.

Definition C.1 (Elicitability). A functional $\mathfrak{T} : \mathcal{M} \to A$, $A \subset \mathbb{R}^k$, is called k elicitable on $\widetilde{\mathcal{M}} \subseteq \mathcal{M}$ if there exists a measurable function $S : A \times \mathbb{R} \to [0, \infty]$—called a strictly consistent scoring function—if for all $F \in \widetilde{\mathcal{M}}$ and for all $z \in A$,

$$\int S(\mathfrak{T}(F), y)\, dF(y) \leq \int S(z, y)\, dF(y), \tag{C.1}$$

and the equality in (C.1) holds only if $z = \mathfrak{T}(F)$.

In the numerical examples, we consider the family of coherent distortion DRM $\{\rho_t\}_{t \in \mathcal{T}}$, a convex combination of the mean and ES, given in (17). We first illustrate that its static version

$$\rho(Z) := p\, \mathrm{ES}_\alpha(Z) + (1-p)\, \mathbb{E}[X], \qquad Z \in \mathcal{Z}_{T+1}, \tag{C.2}$$

is elicitable. Although the mean is well known to be one elicitable, the ES_α is only jointly elicitable together with VaR_α at the same α level. We recall these well-known results.

Proposition C.1 (Mean (Gneiting 2011)). *Let $\phi : \mathbb{R} \to \mathbb{R}$ be strictly convex with subgradient ϕ', and denote by $\mathcal{M}^\dagger \subset \mathcal{M}$ the class of cdfs with finite mean such that $\int |\phi(y)| \mathrm{d}F(y) < +\infty$ for all $F \in \mathcal{M}^\dagger$. Then,*

$$S_{\mathbb{E}}(z, y) := \phi'(z)(z - y) - \phi(z) + \phi(y), \qquad z, y \in \mathbb{R},$$

is strictly $\mathcal{M}^\dagger$ consistent for the mean.

Proposition C.2 ((VaR, ES) (Acerbi and Szekely 2014, Fissler and Ziegel 2016)). *Let $\alpha \in (0,1)$, $A := \{(z_1, z_2) \in \mathbb{R}^2 : z_1 \geq z_2\}$, and define the scoring functions $S_{VaR,ES} : A \times \mathbb{R} \to \mathbb{R}$ by*

$$S_{VaR,ES}(z_1, z_2, y) = (\mathbb{1}_{\{y \leq z_1\}} - \alpha)(g(z_1) - g(y)) + \Phi'(z_2)\left(z_2 - \frac{1}{1-\alpha} S^+_\alpha(z_1, y)\right) - \Phi(z_2) + \Phi(y),$$

where $S^+_\alpha(z_1, y) = (\mathbb{1}_{\{y \leq z_1\}} - \alpha) z_1 - \mathbb{1}_{\{y \leq z_1\}} y + y$, $\Phi : \mathbb{R} \to \mathbb{R}$ is strictly convex with subgradient Φ', and $g : \mathbb{R} \to \mathbb{R}$ is such that for all $z_2 \in \mathbb{R}$,

$$z_1 \longmapsto g(z_1) - z_1 \Phi'(z_2)/(1-\alpha)$$

is strictly increasing.

Let $\mathcal{M}^\ddagger \subset \mathcal{M}$ be the cdfs with unique α quantile and finite mean and such that $\int |g(y)| dF(y) < +\infty$ and $\int |\Phi(y)| dF(y) < +\infty$ for all $F \in \mathcal{M}^\ddagger$. Then, $S_{VaR,ES}$ is strictly $\mathcal{M}^\ddagger$ consistent for the couple (VaR_α, ES_α).

Next, we show that for $p \in (0,1)$, the risk measure ρ given in (C.2) is three elicitable: that is, jointly elicitable together with VaR_α and ES_α. The following proposition is different from Fissler and Ziegel (2016, corollary 5.4), which states that ρ is four elicitable; specifically, they show that $(\mathrm{VaR}_\alpha, \mathrm{ES}_\alpha, \mathbb{E}, \rho)$ is jointly elicitable.

Proposition C.3 (Mean ES Risk Measure). *For $p \in (0,1)$, let ρ be given in (C.2), and let the assumptions of Propositions C.1 and C.2 be enforced. Then, the function $S_\rho : A \times \mathbb{R}^2 \to [0, \infty]$ given by*

$$S_\rho(z_1, z_2, z_3, y) := S_{VaR,ES}(z_1, z_2, y) + S_{\mathbb{E}}\left(\frac{z_3 - p z_2}{1-p}, y\right) \tag{C.3}$$

is a strictly $\mathcal{M}^\ddagger$-consistent scoring function for the triplet $(VaR_\alpha, ES_\alpha, \rho)$.

Proof. From Propositions C.1 and C.2, the functionals $(\mathrm{VaR}_\alpha, \mathrm{ES}_\alpha)$ and $\mathbb{E}$ are elicitable. Using Fissler and Ziegel (2016, lemma 2.6), we obtain that $(\mathrm{VaR}_\alpha, \mathrm{ES}_\alpha, \mathbb{E})$ is elicitable with consistent scoring function given by

$$S_{\mathrm{VaR,ES},\mathbb{E}}(z_1, z_2, z_3, y) = S_{\mathrm{VaR,ES}}(z_1, z_2, y) + S_{\mathbb{E}}(z_3, y).$$

Next, we apply Osband's *revelation principle*; see, for example, Gneiting (2011, theorem 4). First, define the bijective function $g : \mathbb{R}^3 \to \mathbb{R}^3$ by $g(z_1, z_2, z_3) = (z_1, z_2, pz_2 + (1-p)z_3)^\top$ with inverse $g^{-1}(a_1, a_2, a_3) = \left(a_1, a_2, \frac{a_3 - p a_2}{1-p}\right)^\top$. The revelation principle states that $g(\mathrm{VaR}, \mathrm{ES}, \mathbb{E}) = (\mathrm{VaR}, \mathrm{ES}, \rho)^\top$ is elicitable with scoring function

$$\begin{aligned} S_{\mathrm{VaR,ES},\rho}(z_1, z_2, z_3, y) &= S_{\mathrm{VaR,ES},\mathbb{E}}(g^{-1}(z_1, z_2, z_3)^\top, y) \\ &= S_{\mathrm{VaR,ES}}(z_1, z_2, y) + S_{\mathbb{E}}\left(\frac{z_3 - p z_2}{1-p}, y\right). \end{aligned}$$

Moreover, if the scoring functions $S_{\mathrm{VaR,ES}}(z_1, z_2, y)$ and $S_{\mathbb{E}}(z_3, y)$ are strictly consistent for (VaR, ES) and $\mathbb{E}$, respectively, then $S_{\mathrm{VaR,ES},\rho}(z_1, z_2, z_3, y)$ is strictly consistent for $(\mathrm{VaR}, \mathrm{ES}, \rho)$. □

In our implementations, we make the specific choices of $\phi(z) = z^2$, $g(z) = C$ and $\Phi(z) = -\log(z + C)$, where $C > 0$.

Finally, we need to elicit the conditional cdf $F_{Y|\boldsymbol{X}} : \mathbb{R} \to [0,1]$, defined by $F_{Y|\boldsymbol{X}}(y) := \mathbb{P}(Y \leq y | \boldsymbol{X} = \boldsymbol{x})$, for any $Y \in \mathcal{Z}_{t+1}$, $\boldsymbol{X} \in \mathcal{Z}_t$, and $\boldsymbol{x} \in \mathbb{R}^n$. Cdfs are known to be elicitable with the *continuous ranked probability score*; see, for example, Gneiting and Raftery (2007, equation 20). Here, we recall the key result.

Proposition C.4 (Distribution Function (Gneiting and Raftery 2007)). *Let $\mathcal{M}^\dagger$ be the set of cdfs with finite mean. Then, the scoring function $S_{cdf} : \mathcal{M}^\dagger \times \mathbb{R} \to [0, \infty]$ given by*

$$S_{cdf}(F, y) := \int_{\mathbb{R}} (F(z) - \mathbb{1}_{z \geq y})^2\, dz \tag{C.4}$$

is a strictly consistent scoring function for F_Y. In particular, it holds that

$$\operatorname*{arg\,min}_{F \in \mathcal{M}^\dagger} \mathbb{E}[S_{cdf}(F, Y)]$$

is attained by the cdf $F_Y : \mathbb{R} \to [0,1]$.

Next, we consider the concept of conditional elicitability. Let $(\boldsymbol{X}, Y)$ be a random vector with joint cdf $F_{\boldsymbol{X},Y}$, where Y is a univariate rv with cdf F_Y and $\boldsymbol{X}$ is an n-dimensional random vector with cdf $F_{\boldsymbol{X}}$. Further, let $S : A \times \mathbb{R} \to [0, \infty]$ be a strictly consistent scoring function that elicits the functional $\mathfrak{T} : \mathcal{M} \to A$. Next, define the expected score $\mathfrak{S} : \mathcal{M}_{n+1} \to [0, \infty]$ by

$$\mathfrak{S}(F_{\boldsymbol{X},Y},g) := \int S(g(\boldsymbol{x}),y)\,F_{\boldsymbol{X},Y}(d\boldsymbol{x},dy) = \int \left(\int S(g(\boldsymbol{x}),y)\,F_{Y|\boldsymbol{X}=\boldsymbol{x}}(dy)\right) F_{\boldsymbol{X}}(d\boldsymbol{x}),$$

where $\mathcal{M}_{n+1}$ is the space of cdfs of $(n+1)$-dimensional random vectors and $\mathcal{G} := \{g | g : \mathbb{R}^n \to \mathbb{R}$, with $\mathfrak{S}(F,g) < +\infty\}$. Then, the expected score satisfies

$$\begin{aligned}\mathfrak{S}(F_{\boldsymbol{X},Y},g) &= \int \left(\int S(g(\boldsymbol{x}),y)\,F_{Y|\boldsymbol{X}=\boldsymbol{x}}(dy)\right) F_{\boldsymbol{X}}(d\boldsymbol{x})\\ \text{(by definition 9)} &\geq \int \left(\int S(\mathfrak{T}(F_{Y|\boldsymbol{X}=\boldsymbol{x}}),y)\,F_{Y|\boldsymbol{X}=\boldsymbol{x}}(dy)\right) F_{\boldsymbol{X}}(\mathrm{d}\boldsymbol{x})\\ &= \int S(\mathfrak{T}(F_{Y|\boldsymbol{X}=\boldsymbol{x}}),y)\,F_{\boldsymbol{X},Y}(d\boldsymbol{x},dy)\\ &= \mathfrak{S}(F_{\boldsymbol{X},Y},\mathfrak{T}(F_{Y|\boldsymbol{X}})).\end{aligned}$$

Therefore, $\mathfrak{T}(F_{Y|\boldsymbol{X}})$ minimizes the expected score $\mathfrak{S}(F,g)$ over all functions g. Note that we may also write $\mathfrak{S}(F,g) := \mathbb{E}[S(g(\boldsymbol{X}),Y)]$ with the expectation taken over a probability measure, where $(\boldsymbol{X},Y)$ has cdf $F_{\boldsymbol{X},Y}$. We can approximate the minimizer of this expected score by seeking over a rich parameterized class of functions (such as NNs), and we estimate the expectation using the empirical mean from simulations. This is what we use when eliciting approximations of $\mathfrak{R}_t[\boldsymbol{\theta}_{t:T}]$ and $U_t[\boldsymbol{\theta}_{t:T}]$.

Appendix D. Additional Information on Numerical Implementation

D.1. Parameters Used in Market Model Simulation

This section contains further details on the simulated market model simulation. In particular, Table D.1 specifies the market model parameters, and Table D.2 gives the correlation matrix of the dependence structure.

Table D.1. Market Model Parameters

	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$
κ_i	4	4.5	5	5.5	6
θ_i	0.01	0.0225	0.04	0.0625	0.09
η_i	0.5	0.875	1.25	1.625	2
μ_i	0.05	0.075	0.10	0.125	0.15

Table D.2. Correlation Matrix of the Dependence Structure for the Market Model

	X_1	X_2	X_3	X_4	X_5	v_1	v_2	v_3	v_4	v_5
X_1	1.0	0.3	0.3	0.3	0.3	−0.5				
X_2	0.3	1.0	0.3	0.3	0.3		−0.5			
X_3	0.3	0.3	1.0	0.3	0.3			−0.5		
X_4	0.3	0.3	0.3	1.0	0.3				−0.5	
X_5	0.3	0.3	0.3	0.3	1.0					−0.5
v_1	−0.5					1.0				
v_2		−0.5					1.0			
v_3			−0.5					1.0		
v_4				−0.5					1.0	
v_5					−0.5					1.0

Note. Only nonzero entries are shown.

Table D.3. Average Execution Time (in Seconds) per Outer Iteration with $m_r = 20$ and $m_f = 5$

	$T+1$			
d	2	4	8	16
2	0.71	1.44	3.26	8.05
4	0.73	1.46	3.28	8.83
8	0.75	1.54	3.39	10.10

D.2. Computation Times

Table D.3 shows the average time that the algorithm takes to execute one full outer iteration, which includes $m_r = 20$ iterations for updating the risk-to-go, $m_f = 5$ iterations for updating the conditional cdf, and 1 iteration for updating the strategy. Models were trained on an Intel Xeon central processing unit (CPU) E5-2630 v4@2 .20 GHz (from 2016) with 64 GB of random-access memory (RAM) equipped with an NVDIA TITAN RTX graphic processing units (GPU) (from 2018). As the results show, the timing scales approximately linearly with time steps but increases only marginally with the number of assets. The largest limitation is the size of the memory on the GPUs for computing gradients.

Endnotes

[1] There are a number of alternative approaches to time inconsistencies (see, e.g., Björk et al. 2021); we, however, opt to use time-consistent criteria.

[2] If $F_{Z|\mathcal{F}_t}(\cdot)$ is continuous, then $U_{Z|\mathcal{F}_t}$ is, conditional on $\mathcal{F}_t$, a uniform rv. If $F_{Z|\mathcal{F}_t}(\cdot)$ is discontinuous, define $\widetilde{F}_{Z|\mathcal{F}_t}(z,\lambda) := \mathbb{P}(Z<z|\mathcal{F}_t) + \lambda\,\mathbb{P}(Z=z|\mathcal{F}_t)$. Next, let V be a uniform $\mathcal{F}_{t+1}$-measurable rv that is, conditional on $\mathcal{F}_t$, independent of Z. Then, $\widetilde{U}_{Z|\mathcal{F}_t} := \widetilde{F}_{Z|\mathcal{F}_t}(Z,V)$ is, conditional on $\mathcal{F}_t$, a uniform rv; see Rüschendorf (2013, definition 1.2). For simplicity, we use the notation $U_{Z|\mathcal{F}_t} := \widetilde{U}_{Z|\mathcal{F}_t}$.

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